ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

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Subject Name:	Professional English – I (Lab Integrated)
Subject Code:	EN25C09

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

IMPRESSIVE PERSONALITY: BIOGRAPHICAL NOTE

Elon Musk – entrepreneur, engineer and technology visionary

Elon Musk is a globally recognized entrepreneur, engineer, and technology visionary whose career has been defined by pursuing ambitious solutions to some of humanity's most challenging problems. He is known for leading companies operating across electric vehicles, space exploration, artificial intelligence, brain-computer interfaces, and infrastructure. His approach is centered on challenging conventional assumptions, accelerating technological innovation, and turning ideas that once appeared impractical into commercially viable technologies. Tesla identifies Musk as its co-founder and CEO, while SpaceX has been led by him since its founding in 2002.

Musk began his entrepreneurial career in the emerging internet industry. He co-founded Zip2, an early internet company providing maps and business-directory services, which was acquired by Compaq in 1999. He subsequently co-founded X.com, an online financial-services company that evolved into PayPal and was acquired by eBay in 2002. These early ventures established the foundation for his later strategy of entering technically difficult industries and attempting to fundamentally reshape established markets.

In 2002, Musk founded SpaceX with the long-term objective of reducing the cost of space transportation and enabling humanity to become a multi-planetary civilization. Under his leadership, SpaceX developed the Falcon family of launch vehicles and Dragon spacecraft, while pioneering the practical reuse of orbital-class rockets. The company's achievements include the Falcon 1 becoming the first privately developed liquid-fuel rocket to reach orbit and the subsequent development of reusable Falcon 9 technology. SpaceX is also developing Starship for missions involving the Moon, Mars, and beyond.

Musk's influence extends significantly into sustainable transportation and energy through Tesla. As Tesla's CEO, he has overseen product design, engineering, and manufacturing across electric vehicles, battery systems, and solar-energy products. Tesla's development of vehicles such as the Model S, Model 3, Model X, Model Y, and Cybertruck has contributed to the broader global transition toward electric transportation. The company has also expanded into large-scale energy storage through products such as Powerwall and Megapack.

Beyond transportation and aerospace, Musk has pursued emerging technologies through Neuralink, which focuses on brain-computer interfaces, and The Boring Company, which develops tunneling and transportation infrastructure. His broader portfolio demonstrates a consistent willingness to operate at the intersection of engineering, entrepreneurship, and long-term technological transformation.

Musk's career is ultimately characterized by a high-risk, first-principles approach to innovation. Rather than limiting his ambitions to incremental improvements, he has repeatedly entered industries where established technologies, costs, and assumptions appeared difficult to challenge. Through his leadership of multiple technology companies, he has become one of the most prominent figures associated with ambitious engineering-driven innovation in the modern technology sector

ONE-PAGE PROFILE

One-page professional profile – Harini G

HARINI G

Aspiring Cloud & AI Engineer | AI • Cloud Computing • Software Engineering

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GitHub: https://github.com/harini18g-byte

PROFESSIONAL SUMMARY

I am Harini, currently pursuing B.E. Computer Science and Engineering (CSE) at St. Joseph's College of Engineering. I am an aspiring technology professional interested in Artificial Intelligence, Cloud Computing, Software Engineering, Web Development, and App Development.

I have a basic foundation in Python and am currently strengthening my programming fundamentals through C programming. My career goal is to become a Cloud AI Engineer / AI-focused Software Engineer and pursue opportunities in leading multinational technology companies. I am passionate about learning emerging technologies and applying my knowledge through practical projects.

CAREER OBJECTIVE

To build strong technical and problem-solving skills throughout my engineering journey, specialize in AI and Cloud Computing, and gain practical experience through projects, internships, hackathons, certifications, and online learning. My long-term goal is to contribute to innovative technology solutions in a global organization.

TECHNICAL SKILL SET

- Programming: C, Python
- Core Skills: Programming Fundamentals, Problem Solving, DSA – Beginner Level
- Areas of Interest: Artificial Intelligence, Cloud Computing, Machine Learning, Web Development, App Development
- Tools & Platforms: Git, GitHub, HackerRank, LeetCode
- Current Learning: C Programming and Computer Science Fundamentals

PRACTICAL PROJECTS & HANDS-ON LABS

I have prior experience developing a tour-themed project during my school practicals. Currently, I am building my programming knowledge through C programming exercises and college laboratory work.

Going forward, I plan to develop projects involving software development, AI, and Cloud Computing, gradually progressing from small applications to advanced real-world solutions.

VALUE-ADDED CERTIFICATIONS & ONLINE TRAINING

- NPTEL / SWAYAM:Fundamentals of Computing – Current Course
- HackerRank: Programming and problem-solving practice
- LeetCode: DSA and coding practice
- GitHub: Project development and portfolio building
- Future learning and certifications in AI, Machine Learning, and Cloud Computing

PROFESSIONAL & INTERPERSONAL STRENGTHS

- Curious and eager to learn new technologies
- Creative and interested in innovative solutions
- Adaptable to new tools and programming languages
- Developing strong problem-solving abilities
- Goal-oriented and committed to continuous learning
- Interested in teamwork, collaboration, and technical activities
- Working toward clear and confident communication

FOUR-YEAR ROADMAP

Year 1: Build strong foundations in C, Python, DSA, and Computer Science, while exploring AI and Cloud Computing.

Year 2: Develop core CS and AI/Cloud skills, build projects, and participate in hackathons.

Year 3: Specialize in AI + Cloud Computing, gain internship experience, and strengthen interview preparation.

Year 4: Complete a strong final-year project, build my professional portfolio, and prepare for placements and global technology opportunities.

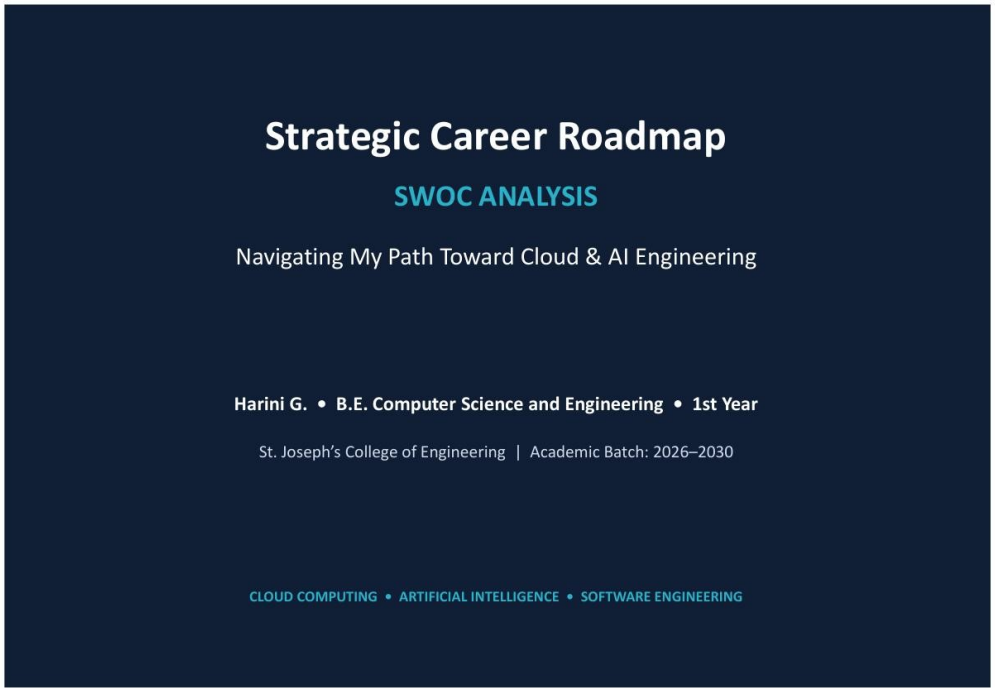
FUTURE VISION

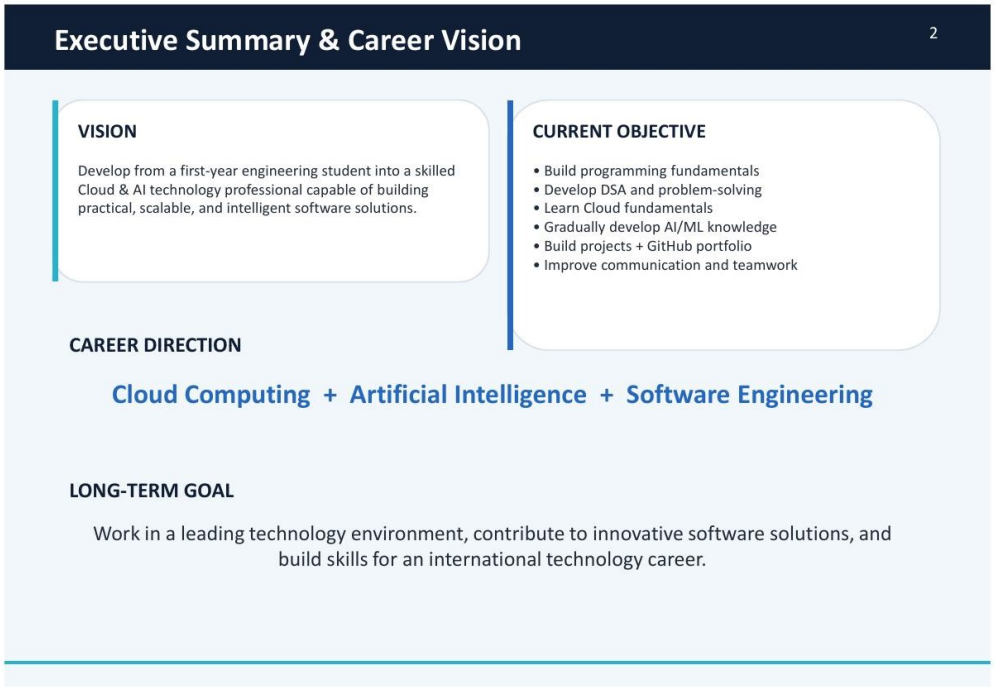
My vision is to become a skilled Cloud AI Engineer / AI-focused Software Engineer with strong technical and practical abilities. I aim to continuously learn, build meaningful projects, gain industry experience, and contribute to innovative technology solutions on a global scale.

"My goal is to transform my interest in technology into practical skills and meaningful solutions."

SWOC PRESENTATION

Strategic Career Roadmap: SWOC Analysis (slides)





Overview of the SWOC Matrix

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STRENGTHS

Curiosity • Creativity • Ambition
Persistence • Tech interest • Adaptability

WEAKNESSES

Beginner programming • Limited projects
Developing DSA • Confidence • Time management

OPPORTUNITIES

Cloud + AI growth • Online learning
Hackathons • Internships • Open source

CHALLENGES

Rapid change • Competition • Academic balance
Many technologies • Continuous skill development

Strengths: Core Foundations

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Early Career Direction

Cloud Computing, AI, and Software Engineering are areas I want to explore.

Curiosity About Technology

I enjoy understanding how software and modern technologies work.

Creative Thinking

I prefer developing unique project ideas rather than only basic exercises.

Willingness to Learn

I am willing to learn difficult concepts and build understanding gradually.

Long-Term Ambition

I want to develop technically and professionally throughout four years.

Adaptability

I am open to new languages, tools, cloud platforms, and emerging technologies.

Strengths: Personal & Learning Qualities

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Growth Mindset

Skills improve through practice, mistakes, and continuous learning.

Persistence

I try to understand difficult concepts instead of immediately giving up.

Self-Directed Learning

I want to learn beyond the syllabus through NPTEL/SWAYAM and HackerRank.

Project-Oriented Thinking

I want to convert learning into practical projects.

Willingness to Improve

I am aware of my weaknesses and want to actively work on them.

Future-Oriented Approach

I want to use four years strategically for skills, projects, experience, and portfolio.

Weaknesses: Technical & Academic Gaps

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Beginner Programming Level

Prior Python exposure; currently strengthening the programming foundation through C.

Developing DSA Skills

Data Structures and Algorithms knowledge is at the foundation-building stage.

Limited Cloud Knowledge

Interested in Cloud, but practical infrastructure and service knowledge is developing.

Limited AI Experience

Beginning the journey toward machine learning and AI development.

Limited Real-World Experience

As a first-year student, professional/industry experience is still limited.

Advanced Systems Exposure

Networking, databases, OS, distributed systems, containers, and cloud architecture need further study.

Weaknesses: Personal & Experience Gaps

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Confidence

Track progress and focus on personal improvement.

Fear of Failure

Treat mistakes as learning opportunities.

Time Management

Use weekly planning and realistic priorities.

Limited Portfolio

Build and document projects progressively across four years.

Need for Consistency

Establish a sustainable weekly learning routine.

CORE PRINCIPLE

Progress matters more than comparison.

Opportunities: Technology & Academia

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Cloud Computing

Learn computing, storage, networking, databases, security, and deployment.
Utilize free cloud tiers (AWS Free Tier / Google Colab) to train models cost-effectively.

Artificial Intelligence

Explore machine learning, intelligent applications, automation, and AI software.

NPTEL / SWAYAM

Use structured online courses to supplement the college curriculum.

Coding Platforms

Use HackerRank to strengthen programming and problem-solving.

College Technical Activities

Use hackathons, symposiums, clubs, workshops, and events for experience.

Academic Projects

Apply programming, AI, and cloud concepts through practical projects.

Opportunities: Portfolio & Professional Development

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GitHub Portfolio

Build a public portfolio of programs, projects, documentation, and learning progress.

AI + Cloud Projects

Combine both interests to create cloud-based intelligent applications.

Internships

Gain real-world development exposure in later years.

Hackathons

Develop teamwork, creativity, problem-solving, rapid development, and presentation skills.

Open-Source Learning

Study real projects and eventually contribute to professional communities.

Professional Networking

Connect with developers, students, mentors, and technology communities.

Challenges: Industry & Career Dynamics

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Rapid Technology Evolution

AI, cloud platforms, frameworks, and development tools change quickly.

High Competition

Many CSE students are developing programming and technology skills.

Experience Gap

Industry roles expect practical skills alongside academic qualifications.

Multiple Technology Choices

Trying everything at once can lead to lack of focus.

Changing Industry Requirements

Expected skills can change over time.

Portfolio Expectations

A strong portfolio requires consistent development, documentation, and improvement.

Challenges: Personal & Academic

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Academic Workload

Balance engineering subjects with technical development.

Time Pressure

Manage coding, projects, courses, college activities, and responsibilities.

Maintaining Consistency

Long-term development requires regular practice.

Technical Complexity

Cloud and AI become progressively more advanced.

Avoiding Overload

Develop depth instead of learning too many technologies at once.

Confidence During Comparison

Focus on personal progress rather than comparing beginnings.

Strategic Action Plan: Weakness Mitigation

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PROGRAMMING GAP

C fundamentals → Python development → regular practice → DSA

CONFIDENCE GAP

Track completed work → participate → present projects → focus on progress

PORTFOLIO GAP

Create GitHub → upload meaningful projects → document learning → scale complexity

TIME-MANAGEMENT GAP

Weekly targets → prioritize core skills → fixed coding time → avoid overload

Goal: turn weaknesses into measurable development areas.

Strategic Action Plan: Using Opportunities

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BUILD THE FOUNDATION FIRST

C → Python → DSA → Linux → Networking → Cloud Fundamentals

CONNECT AI WITH CLOUD

Python → AI/ML Fundamentals → Cloud Services → AI + Cloud Projects

BUILD A PRACTICAL PORTFOLIO

Problem definition • Technology • Development process • Result • Documentation • GitHub repository

GAIN EXPERIENCE

Hackathons • Symposiums • Coding competitions • Workshops • Internships • Project collaborations

CONTINUOUS IMPROVEMENT | Review skills regularly and adjust the roadmap as interests and knowledge develop.

My 4-Year Milestone Matrix

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YEAR 1 2026–27 Build the Foundation	YEAR 2 2027–28 Build Technical Depth	YEAR 3 2028–29 Build Industry Experience	YEAR 4 2029–30 Professional Readiness
C, Python, DSA, Networking & Cloud Basics TARGET: GitHub portfolio + first Cloud project	Advanced DSA, Linux, DBMS, Cloud & AI/ML Fundamentals TARGET: AI-powered Cloud application	Advanced Cloud, AI/ML, Docker, APIs & CI/CD TARGET: Production-style AI + Cloud project + internship	System Design, Advanced AI/Cloud & Placement Preparation TARGET: Major capstone + industry-ready portfolio

FOUNDATION → DEPTH → EXPERIENCE → READINESS


CONCLUSION & KEY TAKEAWAYS
My SWOC Strategy
USE STRENGTHS • FIX WEAKNESSES • SEIZE OPPORTUNITIES • MANAGE CHALLENGES
My four-year engineering journey is an opportunity to transform myself from a beginner CSE student into a technically capable Cloud & AI professional.
LEARN → PRACTICE → BUILD → IMPROVE → CONTRIBUTE
"Learn continuously. Build boldly. Improve every day."
THANK YOU

WRITE AND IMPROVE

Write & Improve – learning experience and score sheets

WRITE & IMPROVE – MY LEARNING EXPERIENCE

English Writing Practice and Skill Development

Introduction

Write & Improve is an online English writing practice tool developed by the University of Cambridge. It provides writing tasks at different levels and allows learners to write, check their work, and make improvements. My experience with the platform gave me an opportunity to practice written English and develop my ability to express ideas in a clear and organized way.

My Experience with Write & Improve

My experience with Write & Improve involved choosing writing activities from the workbooks and completing the tasks by writing my own answers. The activities encouraged me to think about the topic, arrange my ideas, and express them using suitable words and sentences. This helped me practice English writing beyond the usual classroom exercises.

The platform allows learners to submit their writing and receive an estimated level linked to the Common European Framework of Reference for Languages (CEFR). It also highlights areas that may need attention. Reviewing my writing and making changes helped me become more aware of how I use grammar, vocabulary, spelling, and sentence structure.

The process encouraged me to review my work rather than finish a task after the first attempt. By revisiting my writing, I could consider whether my ideas were clear, whether my sentences were organized properly, and whether I had used words appropriately. This made writing practice more active and helped me take greater responsibility for my learning.

What I Gained from the Experience

- Writing confidence: I became more comfortable expressing my thoughts in written English.
- Organization of ideas: I practiced arranging points in a logical order and keeping my writing focused on the topic.
- Grammar awareness: I became more attentive to sentence formation, tenses, articles, prepositions, and punctuation.
- Vocabulary development: I practiced choosing words and expressions that fit the meaning I wanted to communicate.
- Spelling and sentence clarity: Reviewing my writing helped me notice places where wording or spelling could be improved.
- Independent learning: I learned to review my own work, make revisions, and continue practicing without depending entirely on classroom guidance.

Areas It Helped Me Improve

- Paragraph writing: Developing a clear beginning, middle, and ending for a paragraph or short response.
- Sentence structure: Writing complete sentences and connecting ideas more smoothly.
- Grammar and punctuation: Paying closer attention to correctness and readability.
- Vocabulary use: Selecting words suitable for the topic and purpose.
- Task understanding: Reading the question carefully and keeping the response relevant.

- Editing skills: Rereading and revising my work instead of submitting the first draft immediately.

How the Platform Supported My Learning

Write & Improve follows a practice cycle: choose a task, write a response, check it, review the areas that may need attention, and make changes. Its workbooks include tasks for Beginner, Intermediate, and Advanced learners, as well as options such as Business English and Just for Fun. This range allows learners to practice different styles and topics of writing.

The platform also provides a progress graph so learners can compare their current work with earlier attempts. This can help make progress easier to recognize over time. The estimated CEFR level gives an additional reference point for understanding writing development, although it should be treated as guidance rather than a complete judgement of every aspect of a response.

What I Learned About Myself as a Writer

Using the platform helped me understand that good writing involves more than knowing grammar rules. I need to understand the question, organize my thoughts, choose suitable vocabulary, and make sure that my sentences communicate the intended meaning. Reviewing a written answer can reveal areas that may not be obvious while writing the first draft.

I also learned that writing improves through practice and revision. A first attempt may communicate the main idea, but reviewing it can help make the message clearer and more organized. This encouraged me to be more patient with the writing process and to treat mistakes as opportunities to learn.

How It Can Help Me in the Future

- Academic assignments: Writing clearer answers, paragraphs, reports, and project descriptions.
- Emails and formal communication: Organizing messages and choosing an appropriate tone.
- Presentations and reports: Preparing written content that is easy for others to understand.
- Examinations: Practicing structured responses and managing ideas within a task.
- Placements and career development: Preparing professional emails, statements, and other written communication.
- Everyday communication: Expressing opinions, explaining information, and sharing ideas more effectively.

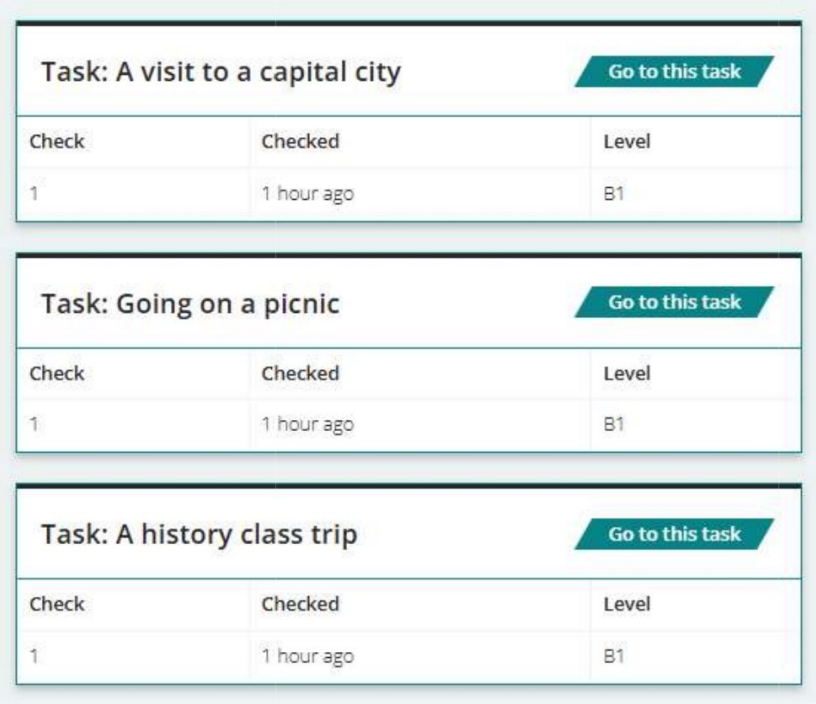
Overall Reflection

Overall, Write & Improve gave me a structured way to practice English writing. Completing workbook tasks encouraged me to develop ideas, write them clearly, and review my work for possible improvements. The platform helped me focus on grammar, vocabulary, sentence structure, organization, and editing as connected parts of writing.

This experience also highlighted the value of regular practice and self-review. By continuing to write and revise different types of responses, I can gradually become a clearer, more confident and more independent writer.

Conclusion

My experience with Write & Improve was useful because it made English writing practice interactive and self-directed. It helped me understand the importance of organizing ideas, using language carefully, and reviewing my work before considering it complete. These skills can support my academic studies, professional communication, and future career.



Task: A visit to a capital city — Go to this task

Check	Checked	Level
1	1 hour ago	B1

Task: Going on a picnic — Go to this task

Check	Checked	Level
1	1 hour ago	B1

Task: A history class trip — Go to this task

Check	Checked	Level
1	1 hour ago	B1

SPEAK AND IMPROVE

Speak & Improve – learning experience and score sheets

SPEAK & IMPROVE – MY LEARNING EXPERIENCE

English Speaking Practice and Skill Development

Introduction

Speak & Improve is an online English speaking practice platform developed as a research project from the University of Cambridge. I used the platform to practice speaking English by answering questions and completing speaking activities. The experience gave me an opportunity to practise expressing my ideas in English and become more comfortable with speaking in a structured online environment.

My Experience with Speak & Improve

My experience with Speak & Improve involved listening to questions, preparing my thoughts, and answering them by speaking into the platform. This was different from simply writing answers because I had to express my ideas verbally and communicate them clearly. The speaking activities encouraged me to think in English and respond in a natural way.

One of the useful features was the ability to listen to my recorded answers and try speaking again. This helped me become more aware of how I communicate and encouraged me to practice the same type of speaking activity more than once. The platform also provides an automatic grade, which gives learners an indication of their speaking ability using the internationally recognized CEFR scale.

The experience helped me understand that improving spoken English requires regular practice. Instead of focusing only on grammar rules, I had to actually use English while speaking. This made the learning process more practical and helped me become more comfortable expressing my thoughts.

What I Gained from the Experience

- Speaking confidence – I became more comfortable speaking in English and expressing my thoughts aloud.
- Better fluency – Regular speaking practice helped me work towards speaking more smoothly and continuously.
- Clearer expression – I learned to organize my ideas before and while speaking so that my answers were easier to understand.
- Pronunciation awareness – Listening to my own recorded responses helped me become more conscious of how I pronounce words.
- Quick thinking – Answering speaking questions encouraged me to think about ideas and respond in English without depending completely on written preparation.
- Independent learning – The online format allowed me to practice speaking on my own and take responsibility for improving my communication skills.

Areas It Helped Me Improve

- Speaking and fluency: I practiced producing spoken English rather than only reading or writing it.
- Vocabulary usage: Speaking activities encouraged me to use words and expressions actively while communicating ideas.
- Grammar in real communication: I became more aware of sentence formation while speaking instead of studying grammar only as written rules.

- Pronunciation and clarity: Recording and listening to my answers helped me pay attention to how clearly I spoke.
- Confidence: Repeated speaking practice helped reduce hesitation and made it easier to communicate in English.
- Communication skills: I practiced answering questions, explaining ideas, and keeping my response focused.

How the Platform Supported My Learning

Speak & Improve follows a simple practice cycle: I answer a speaking question, receive an automatic grade, listen to my answer, and practice again. According to the platform, its speech technology can assess spoken English and provide a grade on the CEFR scale. This makes the activity more structured than simply speaking without any way to monitor progress.

The platform also allows users to create a profile so that progress can be saved and practice can continue across devices. This supports regular learning and makes speaking practice easier to continue over time.

What I Learned About Myself as a Speaker

Using the platform made me more aware of the difference between knowing English and using English confidently. I may understand a grammar rule or know the meaning of a word, but speaking requires me to use that knowledge quickly. The speaking activities helped me identify areas where I needed more practice, especially in organizing thoughts and expressing them clearly.

I also learned the importance of repeated practice. Speaking becomes easier when I practice regularly rather than waiting until I need to speak in an important situation. The experience encouraged me to treat communication as a skill that can be developed through consistent practice.

How It Can Help Me in the Future

- Academic presentations: It can help me explain ideas more clearly during classroom presentations and seminars.
- Group discussions: Speaking practice can make it easier to participate actively and express opinions in discussions.
- Interviews and placements: Confidence in spoken English can help me answer questions clearly during interviews.
- Professional communication: Better speaking skills can support communication with teachers, teammates, colleagues, and future employers.
- Public speaking: Regular practice can help reduce hesitation and improve the ability to speak for longer periods.
- Everyday communication: Greater confidence can make conversations in English more natural and comfortable.

Overall Reflection

Overall, my experience with Speak & Improve was useful because it gave me a practical way to practice English speaking. Instead of concentrating only on written exercises, I had to listen, think, speak, and express my ideas verbally. The ability to listen to my own responses and practice again encouraged me to become more aware of my speaking habits.

The experience helped me understand that confidence, fluency, pronunciation, vocabulary, and clear expression improve through regular practice. It also encouraged me to become a more independent learner and to use digital tools as part of my communication development.

Conclusion

Speak & Improve helped me turn English speaking practice into a regular and practical learning activity. Through speaking tasks and repeated practice, I gained greater confidence and awareness of my communication skills. I believe the skills developed through this experience will be useful for my studies, presentations, interviews, placements, and future professional life.



SUMMARY OF LECTURE NOTES

NPTEL: Enhancing Soft Skills and Personality

Week 1: Foundations of Soft Skills, Perception, and Personal Development

The course introduces soft skills and personality development as abilities that improve through learning, practice, and behavioral change. Important areas include self-awareness, self-confidence, mindset, emotional balance, stress management, coping with failure, patience, trustworthiness, perceptiveness, and spiritual intelligence. Perception affects how people interpret the same situation differently, so empathy and understanding others are essential. Conflict is a natural disagreement that can be managed through communication, negotiation, mediation, and arbitration to achieve win-win solutions. Stress occurs when situational demands exceed an individual's ability to handle them; eustress can improve performance, while distress reduces it. Habits develop through a cycle of stimulus/thought → action → reward/punishment → habit → character, and the Zeigarnik effect explains the mental tension caused by unfinished tasks. Active listening is a purposeful process involving reception, selection, organization, assimilation, interpretation, evaluation, and response. Effective communication also requires proper telephone etiquette, patience, clear speech, empathy, and professional behavior.

Soft Skills → Perception → Self-Management → Conflict → Stress → Habits → Zeigarnik Effect → Listening → Telephone Etiquette.

Week 2: Understanding Conflict, Stress, and Behavioral Patterns

Lectures 6–10 focus on time management, procrastination, and assertiveness. Time management begins with understanding that people perceive time differently; some follow a monochronic approach with fixed schedules, while others follow a polychronic approach that is more flexible. Your attitude toward time reflects your personality, so you should log your time, identify time leakage, control distractions, and work during your peak/biological time. Efficient time use means focusing on high-value activities, planning lifetime goals, breaking them into achievable goals, writing and prioritizing them, and avoiding unnecessary interruptions.

Procrastination means delaying or postponing an important action, creating a gap between what is intended and what is actually done. Parkinson's Law states that work expands to fill the time available for its completion. Procrastination may result from fear, low confidence, laziness, perfectionism, addictions, negative thinking, feeling overwhelmed, or Peter Pan syndrome (emotional/social immaturity). It can cause poor performance, missed opportunities, stress, and regret. To overcome it, break large tasks into small pieces ("eat the elephant bite by bite"), do the difficult/high-value task first ("eat the frog"), use Kaizen/small daily improvements, create a positive work environment, set and declare goals, reward progress, and take action instead of waiting for perfection.

Assertiveness means expressing your rights and opinions without violating the rights of others, creating a win-win situation. It means saying "yes" when you genuinely want to say yes and "no" when you genuinely want to say no, rather than complying with others just to make them happy. Choosing growth instead of staying in a comfort zone is an important part of assertiveness.

Time Perception → Monochronic/Polychronic → Time Leakage → Peak Time → High-Value Activities → Parkinson's Law → Procrastination → Peter Pan Syndrome → Eat the Frog → Kaizen → Zeigarnik Effect → Assertiveness → Win-Win → Say No.

Week 3: Time Management, Procrastination, and Assertive Behaviour

Assertiveness means expressing your rights, opinions, and feelings directly and honestly without violating others' rights, creating a win-win situation. A passive person avoids speaking up, often agrees even when they disagree, and feels anxious or inferior; an aggressive person forces their

opinions, violates others' rights, and may humiliate them. An assertive person is confident, respectful, empathetic, and calm. Assertiveness develops through constant practice, reality checking, changing self-defeating thoughts, making independent decisions, removing self-doubt, and learning to say "no" without guilt.

Negative emotions, especially anger, are natural human emotions, but excessive anger can distort thinking, damage relationships, and reduce emotional intelligence. Anger can be controlled by delaying the reaction, taking deep breaths, staying silent, thinking about consequences, exercising, writing feelings down, forgiving, apologizing when necessary, and responding calmly rather than aggressively.

Positive thinking helps convert negative energy into motivation, confidence, and opportunities for success. Negative thinking creates doubt and prevents action, while a positive mental attitude encourages goal achievement and self-development. To develop positive thinking, create a strong desire and determination to change, introspect, identify and remove limitations, develop a positive self-image, associate with positive people, observe your thoughts, and control your mind.

Assertiveness → Passive vs Aggressive → Win-Win → Saying No → Self-Doubt → Negative Emotions → Anger Control → Emotional Intelligence → Positive Thinking → Self-Introspection → Positive Self- Image → Determination.

Week 4: Emotional Balance, Positive Thinking, and Interpersonal Skills

People skills are important for maintaining good personal and professional relationships. People may dislike someone because of qualities such as ego, self-centeredness, aggression, arrogance, negativity, pessimism, and lack of empathy. Since humans are social beings, it is important to identify and eliminate such qualities through self-analysis, responsibility, and positive change.

To become likeable, develop qualities opposite to these negative traits: selflessness, compassion, kindness, empathy, optimism, encouragement, appreciation, giving credit to others, politeness, trustworthiness, dependability, honesty, and keeping promises. The Socratic triple filter suggests checking whether information is true, good, and useful before accepting or spreading it.

Attractiveness goes beyond physical appearance and includes physical, mental, emotional, and spiritual qualities. Physical attractiveness involves personal hygiene, fitness, proper food, sufficient sleep, graceful body language, and a unique "signature personality." Mental and emotional attractiveness involve intellectual charm, interesting ideas, remembering people's important details, self-respect, and being genuine. Spiritual attractiveness involves mindfulness, living in the present, deeper connections, self-awareness, self-actualization, helping others without expectations, and "paying it forward."

People Skills → Self-Analysis → Ego/Aggression/Negativity → Empathy → Likeability → Socratic Triple Filter → Trustworthiness → Keeping Promises → Attractiveness → Physical/Mental/Emotional/Spiritual → Signature Personality → Mindfulness → Self-Actualization → Help Without Expectations.

Week 5: English Proficiency, Grammar, and Effective Language Use

These lectures mainly focus on English skills and common errors in everyday communication. Important grammar and usage rules include using elder/older correctly, married to (not married with), house for a building and home for a place of belonging, extended family instead of the Indianism "joint family," and relationship for connections between people. Use since for a starting point and for for duration; ago refers to time before the present, while before is generally used with a past reference.

The lectures also emphasize avoiding literal translation from Indian languages into English and learning correct collocations (words that conventionally occur together). Important distinctions

include weather vs climate, above vs over, which vs that, a/an based on sound, and correct expressions such as get ahead, call back, at the top, and despite/in spite of.

Grammar accuracy also requires subject–verb agreement, correct pronouns, articles, and countable/uncountable nouns. Collective nouns can be singular or plural depending on meaning; uncountable nouns such as information, advice, machinery, furniture, milk, and rice generally take singular verbs. With either…or/neither…nor, the verb agrees with the subject closest to it. Words such as politics, physics, mathematics, and ethics look plural but generally take singular verbs.

Common Errors → Indianisms → Correct Usage → Collocations → Articles → Subject–Verb Agreement → Countable/Uncountable Nouns → Since/For → Ago/Before → Weather/Climate → Above/Over → Which/That → Pronouns → English Accuracy.

Week 6: Humour, Financial Awareness, Health, and Well-Being

Humour is an important communication and personality-development tool that creates positivity, reduces worries, increases happiness, and makes difficult situations easier to handle. It is also contagious and can help people connect better. In the workplace, humour can reduce stress and conflicts, improve creativity and problem-solving, increase productivity, break the ice, strengthen relationships, and make work more enjoyable. However, humour should be used with empathy and proper timing; serious or sensitive situations require seriousness. A playful attitude can reduce the feeling of being overwhelmed and encourage a growth mindset.

Money and personality are connected because a person's character is reflected not only by how much money they have, but by how they handle money. Treating rich and poor people differently can indicate a lack of emotional balance or maturity. Returning borrowed money on time, being trustworthy, and maintaining integrity build credibility and self-esteem.

Managing money requires budgeting, maintaining a balance sheet, tracking income and expenses, controlling unnecessary spending, saving regularly, avoiding greed and impulsive purchases, comparing prices, and buying only necessary/high-value items. Wealth should be created through hard work, knowledge, discipline, and proper money management, not simply luck. A healthy attitude is “detached attachment”—value money but do not allow wealth or loss to control your emotional balance.

Humour → Happiness → Stress Reduction → Conflict Resolution → Creativity → Productivity → Break the Ice → Empathy → Playful Attitude → Money Personality → Integrity → Trustworthiness → Budgeting → Balance Sheet → Savings → Avoid Greed → Impulsive Purchases → Wealth Creation → Detached Attachment.

Week 7: Relationships, Professional Etiquette, and Mind Management

Health is a state of complete physical, mental, and social well-being, not merely the absence of disease. Good health improves energy, confidence, discipline, reliability, and personality. Regular exercise such as walking, jogging, cycling, swimming, or sports keeps the body fit; consistency is more important than intensity. Yoga and meditation help control stress, while morning exercise can provide sunlight and vitamin D.

Diet and sleep are essential parts of health. A balanced diet should include regular meals, fruits and vegetables, proper carbohydrates, proteins and fats, and sufficient clean water; unhealthy, preserved, sugary, and highly processed foods should be limited. Adequate sleep (about 7–8 hours) is important for physical and mental well-being. Avoid heavy late-night meals, long afternoon naps, and electronic gadgets in bed, and maintain a clean, comfortable sleeping environment.

Love is a positive emotion involving understanding, optimism, goodness, truth, faith, fairness, friendship, responsibility, caring, compassion, and respect. Real love is unconditional, patient, kind, trustworthy, and not based on control or possession. It should support personal growth rather than

restrict it. In relationships, effective communication requires understanding body language, psychological timing, mutual respect, trust, independence, and conflict resolution. In professional environments, love should never disturb work, studies, discipline, or workplace decorum.

Health → Physical/Mental/Social Well-being → Exercise → Yoga & Meditation → Consistency → Balanced Diet → Sleep → 7–8 Hours → Love → Unconditional → Compassion → Respect → Trust → Personal Growth → Body Language → Psychological Timing → Professional Decorum.

Week 8: Memory Development, Cognitive Skills, and Environmental Awareness

Business etiquette means the professional manners expected in business and social situations, such as being polite, punctual, respectful, clear in communication, not interrupting, avoiding unethical actions, maintaining confidentiality, and behaving professionally. Good etiquette develops through continuous observation and practice.

Mind management means organizing and focusing thoughts instead of allowing the mind to become scattered. An organized and focused mind improves energy, happiness, productivity, and decision- making. Use positive self-talk/autosuggestion, focus on self-improvement rather than comparing yourself with others, collect relevant information before making important decisions, and allow the mind to process information before deciding.

Memory improves through attention, association, repetition, imagination, and practice. To remember names, hear them carefully, ask for spelling/pronunciation if necessary, repeat the name at least three times, and associate the name with its meaning or a distinctive facial feature. Adequate sleep, proper diet, and stress management also support memory. Memory aids such as lists, notes, puzzles, chess, and learning a new language can improve recall.

Environmental care is essential because humans depend on nature for air, water, food, energy, and living space. Excessive urbanization, modernization, and exploitation of natural resources damage ecosystems. Excessive fossil-fuel use increases CO_2 and the greenhouse effect, contributing to global warming, climate change, melting glaciers, rising sea levels, floods, droughts, and climate displacement. Therefore, humans must reduce resource exploitation and adopt sustainable and environmentally responsible lifestyles.

Business Etiquette → Professionalism → Integrity → Mind Management → Focus → Autosuggestion → Positive Thinking → Effective Thinking → Memory → Attention → Association → Repetition → Sleep/Diet → Environment → Ecosystem → Natural Resources → CO_2 → Greenhouse Effect → Global Warming → Climate Change → Sustainability.

PROJECT PROPOSAL: IDEATION

LifeLink – Smart Emergency Safety & Tracking System | One Tap. One Route. One Lifeline.

01 Introduction

LifeLink is a personal safety mobile application that combines journey planning, emergency access, vehicle details, and time-based safety check-ins in one place. It helps people prepare before travelling and respond faster when a journey becomes worrying or an emergency occurs.

Problem addressed

During a stressful situation, users may lose time switching between maps, contacts, alarms, and messaging tools. LifeLink proposes one simple workflow: plan a route, set an expected arrival time, keep emergency options accessible, and confirm safe arrival.

Main objective

To create a simple, privacy-conscious safety prototype connecting route navigation, an SOS alarm, emergency numbers, vehicle trip details, and a journey timer.

PLAN	CHECK IN	RESPOND
Choose a destination and route.	Set arrival time and trusted contacts.	Access SOS and emergency shortcuts.

02 Proposed Solution and Features

The application is organized around one active journey at a time.

1. **Mapping and Fixed Routes** – Show current location and a route to a destination such as home, college, or a bus stop. Saved routes can be reopened quickly.
2. **Emergency Alarm and SOS** – A prominent SOS control can activate a loud phone alarm, show emergency-call options, and send an alert with available location data to trusted contacts.
3. **Emergency Numbers** – Keep police, ambulance, fire services, and trusted contacts easy to reach. For an India-focused prototype, include 112.
4. **Vehicle Number and Trip Details** – Let users enter a vehicle registration number and basic trip details, then share them with a trusted contact if required.
5. **Time Limits and Safety Check-ins** – Set an expected arrival time. If the timer expires, the app prompts the user to confirm safety and can send a configured alert.
6. **Minimal Data Storage** – Avoid continuous location-history storage by default. Keep trip details only as long as needed for the active session and delete temporary data under clear rules.

Important: A phone alarm does not automatically notify police. Remote alerts depend on permissions, connectivity, device settings, and implemented services. Entering a vehicle registration number does not verify the owner's identity.

03 Working Process

A simple flow makes the concept easy to explain and demonstrate.

STEP 1 — START JOURNEY	STEP 2 — SET SAFETY TIMER
Choose destination, route, and optional vehicle details.	Set expected arrival time and select trusted contacts.
STEP 3 — FOLLOW ROUTE	STEP 4 — CONFIRM ARRIVAL

Keep map and SOS controls accessible during travel.	Tap "I'm Safe" when you reach your destination.

Emergency response

WHEN SOS IS PRESSED	WHEN THE TIMER EXPIRES
The app can activate the configured alarm, show emergency calling options, and send an alert with available location information to trusted contacts.	The app prompts the user to confirm safety. If there is no response, it can send a preconfigured check-in alert to the selected contact.

Example scenario: A student travelling home selects a saved route, enters the cab registration number, and sets a 30-minute check-in timer. On arrival, the student taps "I'm Safe." If the timer expires first, the app requests confirmation and can notify the selected contact if there is no response.

PLAN YOUR JOURNEY • STAY CHECKED IN • GET HELP FASTER

04 Technology Stack and Feasibility

A first version can be built in stages using widely available tools.

TECHNOLOGY	PURPOSE	REASON FOR CHOICE
Flutter	Mobile app interface	One codebase can support Android and iOS.
Google Maps Platform or OpenStreetMap	Maps and route planning	Provides map and route options for the prototype.
GPS / device location services	Current location	Uses phone location with user permission.
Firebase Authentication	Optional user sign-in	Can simplify account setup for a demo.
Firebase Cloud Messaging	Notifications to registered devices	Useful for alerts; setup and testing are required.
Local notifications and timers	Countdown and check-in reminders	Supports the time-limit feature.
Local database	Temporary trip details	Stores only information needed for the active session.

Proposed development timeline

WEEK 1	WEEK 2	WEEK 3	WEEKS 4–5
Design screens and create map/route flow.	Add SOS screen, emergency shortcuts, and timer.	Add trusted-contact alerts and arrival check-in.	Test permissions, false alarms, connectivity, and deletion.

Feasibility: A first-year CSE team can demonstrate the core workflow in approximately 3–5 weeks, depending on experience and available devices. Reliable background tracking and remote alert delivery need real-device testing and careful permission handling.

05 Novelty, Challenges and Conclusion

The central idea is the connected safety journey—not a claim that every feature is entirely new.

What makes LifeLink different

- **Journey-centred safety** – Route, timer, arrival confirmation, and SOS are linked in one flow.

- **Missed check-in response** – A time limit prompts the user to confirm safety and can trigger a configured alert.
- **Vehicle-aware trip details** – Vehicle registration details can be included in a trip summary shared with a trusted contact.
- **Privacy-conscious design** – Location history is minimized by default and temporary trip data can be deleted after the session.

Challenges and practical responses

CHALLENGE	PRACTICAL RESPONSE
Weak internet connection	Keep local alarm and emergency-number shortcuts available; show when remote alerts cannot be sent.
False alarms or missed timers	Use a confirmation prompt, clear cancel option, and user-controlled alert settings.
Privacy and misuse	Use consent, limited permissions, secure handling, and automatic deletion of temporary data.
Background app limitations	Test on real phones and explain battery, operating-system, and notification limitations.

Conclusion

LifeLink combines fixed-route navigation, emergency alarms, quick access to emergency numbers, vehicle details, and time-limited safety check-ins. Users select a destination, set an expected arrival time, and choose trusted contacts. If they do not confirm safe arrival, the app can prompt them and send a configured alert. By focusing on timely check-ins and minimal data storage, LifeLink aims to make everyday journeys safer without encouraging unnecessary surveillance.

LIFELINK – ONE TAP. ONE ROUTE. ONE LIFELINE.

GITHUB PROFILE

GitHub profile README – Harini G

Hi there, I'm Harini G.

Aspiring Cloud & AI Engineer | AI • Cloud Computing • Software Engineering

LinkedIn: https://www.linkedin.com/in/harini-g-067943321/ | GitHub: https://github.com/harini18g-byte | Email: harini18g@gmail.com

B.E. Computer Science & Engineering | 1st Year – Chennai, Tamil Nadu, India

About Me

I'm a first-year Computer Science Engineering student with a growing interest in **Cloud Computing, Artificial Intelligence, and Software Engineering**. I am currently building my technical foundation through programming, problem-solving, Data Structures & Algorithms, Linux, networking, and cloud fundamentals.

- Currently developing my skills in **C and Python**
- Exploring **Cloud Computing and Cloud Technologies**
- Exploring **Artificial Intelligence and Machine Learning**
- Interested in building practical and meaningful software solutions
- Working toward becoming a **Cloud & AI Engineer**
- Continuously learning through academics, projects, coding practice, and online resources
- Interested in combining **AI + Cloud** to create intelligent and scalable applications

Tech Stack & Learning

Domain	Technologies
Programming	C • Python
Computer Science	Data Structures & Algorithms
Cloud Computing	Cloud Fundamentals • Cloud Architecture — Learning
Artificial Intelligence	AI • Machine Learning — Exploring
Operating Systems	Linux — Learning
Networking	Computer Networking — Learning
Tools	Git • GitHub • VS Code
Development	Software Engineering Fundamentals

Career Roadmap

Year 1 — Build the Foundation

C → Python → DSA → Linux → Networking → Cloud Fundamentals

Focus on developing strong programming and computer science fundamentals.

Year 2 — Build Technical Depth

Advanced DSA → DBMS → Cloud → AI/ML Fundamentals

Develop deeper knowledge and begin building more advanced applications.

Year 3 — Build Industry Experience

Advanced Cloud → AI/ML → Docker → APIs → CI/CD → Internship

Focus on real-world development and industry exposure.

Year 4 — Become Industry Ready

System Design → Advanced AI/Cloud → Major Project → Placement Preparation

Build a strong professional portfolio and prepare for an industry career.

Featured Projects

Cloud Computing Projects

I'm currently developing my project portfolio and exploring cloud-based application development.

Areas of interest:

- Cloud-based applications
- Cloud deployment
- Scalable software solutions
- Cloud infrastructure
- Cloud security fundamentals

Artificial Intelligence Projects

I'm exploring Artificial Intelligence and Machine Learning while developing my understanding through practical projects.

Areas of interest:

- Machine Learning
- Intelligent applications
- AI-powered automation
- Data-driven solutions

AI + Cloud Projects

My long-term focus is combining **Artificial Intelligence with Cloud Computing** to develop intelligent, scalable, and practical applications.

Current Goals

- ▱ Strengthen C programming
- ▱ Improve Python
- ▱ Learn Data Structures & Algorithms
- ▱ Improve problem-solving skills
- ▱ Learn Linux fundamentals
- ▱ Understand computer networking
- ▱ Learn Cloud Computing fundamentals
- ▱ Build my first Cloud project
- ▱ Explore AI/ML
- ▱ Build an AI-powered project
- ▱ Build an AI + Cloud project
- ▱ Participate in hackathons
- ▱ Explore open-source contributions

- Build a strong GitHub portfolio
- Prepare for internships

Learning Philosophy

I believe that strong technical skills are developed through **consistent learning, practice, and practical application**.

Learn → Practice → Build → Improve → Contribute

Every project and learning experience is an opportunity to improve my technical and problem-solving abilities.

Certifications & Achievements

I'm currently building my technical foundation. Certifications, hackathon achievements, courses, projects, and other accomplishments will be added here as I earn them.

GitHub Activity

GitHub stats cards (github-readme-stats, streak stats and top languages) for the user harini18g-byte are displayed here on the live profile.

Career Vision

"Learn continuously. Build boldly. Improve every day."

My goal is to grow from a Computer Science Engineering student into a skilled **Cloud & AI Engineer** capable of developing practical, scalable, and intelligent technology solutions.

Connect With Me

I'm interested in connecting with fellow students, developers, mentors, and technology enthusiasts interested in **Cloud Computing, Artificial Intelligence, and Software Engineering**.

LinkedIn: https://www.linkedin.com/in/harini-g-067943321/

GitHub: https://github.com/harini18g-byte

Email: harini18g@gmail.com

⋆**Thanks for visiting my profile!** Cloud • AI • Code • Innovation

LINKEDIN PROFILE

LinkedIn profile – Harini G

HARINI G

CSE Student @ St. Joseph's College of Engineering & EGAMS | Aspiring Cloud & AI Engineer | Learning C, Python, Java & DSA

Greater Chennai Area

Contact: harini18g@gmail.com | www.linkedin.com/in/harini-g-067943321 (LinkedIn)

Summary

I am a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering with a strong interest in software development, Artificial Intelligence, and Cloud Computing. I am currently building my programming foundation in C, Python, and Java while learning Data Structures and Algorithms and improving my problem-solving skills. I am also exploring application development, web development, AI, and cloud technologies to discover how different areas of technology come together to create real-world solutions. I have completed foundational certifications in Data Science and AI and Electronic Systems, and I am continuing to develop my technical knowledge through academic learning, online courses, coding practice, and hands-on projects. My current goal is to become a Cloud &AI Engineer and build a strong foundation in software engineering, cloud technologies, and artificial intelligence. I look forward to gaining practical experience through projects, internships, certifications, and collaborative opportunities throughout my college journey. I believe in continuous learning, curiosity, consistency, and turning what I learn into practical solutions.

Top Skills

- Microsoft Visual Studio Code
- GitHub
- Application Development

Certifications

- DATA SCIENCE AND AI
- ELECTRONIC SYSTEMS

Education

- **Edison G Agoram Memorial School** · (April 2012 - April 2026)
- **St. Joseph's College Of Engineering** – Bachelor of Engineering, Computer Science and Engineering · (August 2026 - May 2030)

INTERCHANGE – CAMBRIDGE ONE EXPERIENCE

English Communication and Lab Activities

Introduction

During this course, our class used the Cambridge One platform for English communication and lab activities. Instead of relying only on textbooks or classroom explanations, we practiced English through online units, exercises, and tests. The platform provided a structured and self-paced way to practise and improve my language skills.

My Experience Working on Cambridge One

My main experience with Cambridge One was completing the exercises provided through the portal during our lab sessions. I worked through the assigned activities step by step and used the practice from the platform to understand my mistakes. The exercises covered different areas of English, so the lab sessions gave me an opportunity to practice what we had learned in class rather than only studying the concepts theoretically.

Working directly on the portal made the learning process more active. I could attempt an exercise, check my answer, identify the area where I made a mistake, and try again. This helped me pay more attention to the details of language use. Repeating difficult activities also helped me become more confident and comfortable with the topics.

The online format also encouraged me to complete the activities regularly and take responsibility for my own progress. The combination of classroom learning and Cambridge One practice made the course more useful because I could immediately apply concepts through exercises.

What I Learnt

- Listening: I learned to focus on the main idea and pick out specific details from recorded conversations, announcements, and talks. Replaying audio helped me catch information I missed the first time and become more comfortable with different accents and speaking speeds.
- Reading: I improved my reading speed and comprehension. I learned to skim for the overall idea, scan for particular information, and guess the meaning of unfamiliar words from context.
- Writing: I learned how to structure emails, paragraphs, and short pieces of writing with a clear beginning, middle, and end. I also became more careful about punctuation, sentence formation, and choosing the right tone.
- Grammar: I revised important topics such as tenses, articles, prepositions, and sentence structure. The exercises helped me revise important grammar topics and practice using them correctly.
- Vocabulary: I learned new words, phrases, and collocations that are useful in academic and professional settings and practiced using them more naturally.
- Study Habits: I learned to manage my time, complete tasks regularly, and learn independently through online practice.

What I Liked About the Cambridge One Platform

- Interactive exercises gave me regular opportunities to practice the topics covered in class.
- The self-paced format allowed me to revisit difficult topics and practice them again.
- Scores and progress tracking helped me understand which areas needed more attention.
- Well-organized units moved from basic to more advanced topics in a logical order.
- Lab sessions provided dedicated time to complete online exercises and practice English instead of learning only through theory.

How the Lab Exercises Helped Me

The Cambridge One exercises helped me connect classroom lessons with practical usage. Rather than simply reading rules or notes, I had to apply them while answering questions and completing activities. This made it easier to recognize how English is used in real situations.

The regular practice also helped me become more careful while reading questions and listening to audio. The exercises gave me an opportunity to practice the topics repeatedly and become more familiar with their correct usage.

Another useful part of the experience was the flexibility of online practice. When I found a topic difficult, I could spend more time on it and repeat the relevant activities. This made my learning more independent and helped me build a habit of practicing consistently.

Skills Developed Through the Experience

- Better listening and understanding of spoken English.
- Improved reading speed, comprehension, and ability to find specific information.
- More confidence in writing clear and organized sentences, emails, and short paragraphs.
- Stronger understanding of grammar through repeated practice.
- Expanded vocabulary for academic and professional communication.
- Better time management, consistency, and independent learning habits.
- Greater confidence in using digital learning platforms for regular practice.

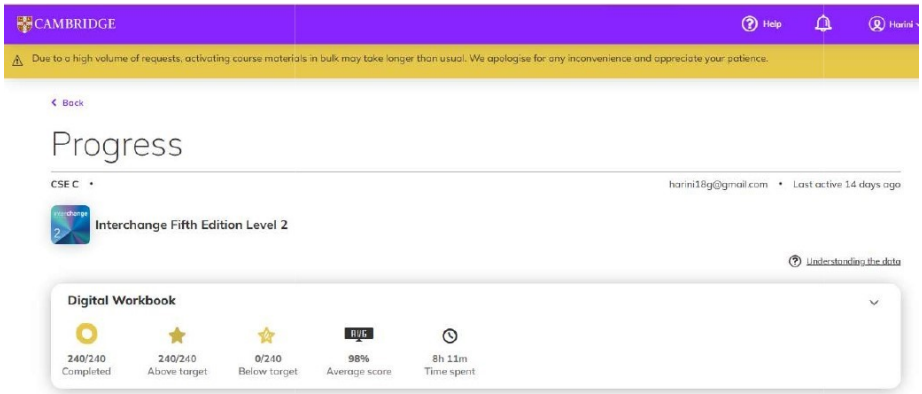
How This Will Help Me in the Future

The skills I developed through the Interchange course and Cambridge One activities will support me in many areas of my academic and professional life. Better communication will help me write clear emails and reports, understand spoken and written information in professional settings, and communicate more confidently with classmates, teachers, and future colleagues.

The practice I gained from completing online exercises will also be useful when preparing for placements and interviews. Being able to understand instructions, communicate ideas clearly, and use English accurately are important skills for academic projects and future workplace situations.

Conclusion

Overall, my experience with Cambridge One made the Interchange course more practical and interactive. Working out the exercises during lab sessions helped me apply the concepts learned in class, identify my mistakes, and improve through repeated practice. The experience strengthened not only my English communication skills but also my confidence, consistency, and ability to learn independently.



CEFR LEVEL PROOF: SELF INTRODUCTION

About Me and My Career – written script with CEFR writing assessment

About Me and My Career

My name is Harini G, and I am a first-year Computer Science and Engineering student at St. Joseph's College of Engineering. I am interested in technology, programming, and learning new things.

My career goal is to become a Cloud AI Engineer. I want to combine Cloud Computing and Artificial Intelligence to create useful and innovative solutions. I aim to improve my programming, problem-solving, and software development skills through projects, certifications, and internships.

In the future, I hope to work for a leading technology company, gain international work experience, and contribute to new developments in technology. I believe that continuous learning, dedication, and hard work will help me achieve my goals.

CEFR Writing Assessment

Overall estimated level: B1+ (Strong Intermediate)

Honest verdict: The writing sample is best assessed as B1+, rather than fully B2. It communicates ideas clearly and logically, uses appropriate vocabulary, and maintains a professional tone. However, the vocabulary and sentence structures are mostly familiar and predictable, with limited variety.

Assessment area	Estimated level
Grammar	B1+
Vocabulary	B1+
Sentence structure	B1
Coherence	B2
Clarity	B2
Professional communication	B1+

Next milestone: B2 **Long-term target:** C1

CEFR LEVEL PROOF: DIALOGUE WRITING (NEURALINK)

Neuralink: A Conversation About the Future – with CEFR writing assessment

One evening, a curious boy named Arjun saw a middle-aged man reading a science magazine in a park. He noticed a picture of a human brain connected to a computer and decided to ask about it.

Arjun: Excuse me, sir. I noticed this picture in your magazine. What is Neuralink?

Man: Neuralink is a neurotechnology company co-founded by Elon Musk. It develops technology that connects the human brain to computers.

Arjun: That sounds interesting! Does it mean people can control computers using their thoughts?

Man: Yes, to some extent. In clinical trials, some participants have learned to move a computer cursor and click using signals from their brains.

Arjun: That's amazing! But how can this technology help people in real life?

Man: It could help people with paralysis use computers and communicate more independently, even when they cannot move their hands.

Arjun: So, this invention is not just about technology but also about improving people's lives!

Man: Exactly, Arjun. Science becomes more meaningful when it helps solve real-world problems.

Arjun: Sir, where is Neuralink located?

Man: Its headquarters is in Fremont, California, in the United States. However, its technology is still experimental, and researchers are continuing to study its safety and effectiveness.

Arjun: Do you think this technology will become common in the future?

Man: It is possible, but we must ensure that it is safe, reliable, and used responsibly.

Arjun looked at the magazine with excitement. He realized that scientific discoveries could open new possibilities and help people overcome difficult challenges.

Moral: Technology becomes truly powerful when it is used to improve human lives.

CEFR Writing Assessment

Overall estimated level: B2 (Upper-Intermediate)

Honest verdict: This story demonstrates B2-level features in its vocabulary, sentence construction, organization, and ability to explain a scientific topic through dialogue. It uses topic-specific words such as “neurotechnology,” “clinical trials,” “paralysis,” “experimental,” and “effectiveness.” The dialogue is easy to follow and develops from a basic question to a discussion of real-world benefits and responsible use.

Assessment area	Estimated level
Grammar	B2
Vocabulary	B2
Sentence structure	B2
Coherence and organization	B2
Clarity	B2
Dialogue writing	B2

Next milestone: C1 – Advanced

MODEL EXAM I – PRACTICE WORKSHEET

EN25C09: Professional English – I (Lab Integrated)

You Choose, We Do It
St. JOSEPH'S COLLEGE OF ENGINEERING
(An Autonomous Institution)
St. Joseph's Group of Institutions
OMR, CHENNAI - 119

PART - A

Answer ALL questions.

Exercise 1: Simple Present Tense (Subject-Verb Concord) Fill in the blanks with the correct form of the verbs given in brackets:

1. Neither the project supervisor nor the lead programmers _____ (recognize / recognizes) the anomaly in the build script.

Answer: recognize

2. Each of the automated sorting arms _____ (require / requires) recalibration after twelve hours of continuous duty.

Answer: requires

3. Either the primary sensor or the auxiliary telemetry units _____ (transmit / transmits) packet data to the ground station.

Answer: transmit

4. The consortium of aerospace contractors _____ (agree / agrees) to adhere to the revised tolerance levels.

Answer: agrees

5. Physics and material science _____ (form / forms) the core theoretical foundation for nanoscale fabrication.

Answer: form

Exercise 2: Simple Past Tense (Trial Reporting) Fill in the blanks with the simple past form of the given verbs:

1. Last month, the metallurgy division _____ (conduct) destructive fatigue testing on the titanium alloy shaft.

Answer: conducted

2. The field engineers _____ (deploy) thirty IoT water-monitoring nodes across the river basin.

Answer: deployed

3. During the pressure overload trial, the relief valve _____ (open) at 150 psi as designed.

Answer: opened

4. The research unit _____ (publish) their initial benchmark results on low-power RISC-V architectures.

Answer: published

5. In the 2025 dry run, the autonomous navigation algorithm _____ (detect) all dynamic road obstacles successfully.

Answer: detected

Exercise 3: Comparative Adjectives (Technical Metrics) Complete each sentence with the appropriate comparative form of the adjective in brackets:

1. Gallium nitride (GaN) semiconductors operate at _____ (high) power efficiency than standard silicon transistors.

Answer: higher

2. Solid-state batteries deliver a _____ (safe) operational threshold under rapid discharge cycles than traditional lithium cells.

Answer: safer

3. The revised compiler generates code that executes _____ (fast) than the legacy unoptimized routine.

Answer: faster

4. Brushless DC motors are considerably _____ (durable) than brushed equivalents due to reduced mechanical wear.

Answer: more durable

5. The liquid cooling loop maintained a _____ (low) core temperature than the dual fan-sink array.

Answer: lower

Exercise 4: Modals of Obligation & Necessity Fill in each blank with an appropriate modal auxiliary (must, should, or have to / has to):

1. Every operator _____ wear anti-static wrist straps before handling open CMOS motherboards.

Answer: must

2. Hazardous chemical waste _____ be neutralized prior to disposal into the industrial drain network.

Answer: must

3. Field technicians _____ log their entry time into the high-voltage testing bay without fail.

Answer: have to

4. To maintain ISO certification, calibration logs _____ be audited biannually by an external inspector.

Answer: must

5. The test rig _____ undergo an emergency stop inspection before initiating the high-speed turbine run.

Answer: must

Exercise 5: Modals of Capability & Possibility Fill in each blank with a suitable modal expressing capability or possibility (can, could, may, might):

1. With the upgraded DSP chip, the system _____ process 4K video streams in real time.

Answer: can

2. If thermal throttling occurs, the processing clock rate _____ drop by as much as 30%.

Answer: may

3. Advanced laser profilometry _____ measure surface roughness down to sub-micron tolerances.

Answer: can

4. Unshielded signal lines _____ pick up electromagnetic interference from nearby induction coils.

Answer: could

5. The modular chassis design _____ accommodate up to four secondary expansion cards.

Answer: can

Exercise 6: Relative Clauses

Combine each pair of sentences into a single coherent sentence using a suitable relative pronoun (who, which, that, whose):

1. (a) A civil engineer designed the suspension cable bridge. (b) She received the national infrastructure award.

Answer: The civil engineer who designed the suspension cable bridge received the national infrastructure award.

2. (a) The spectrometer analyzes infrared emission spectra. (b) It is mounted inside the cleanroom vacuum chamber.

Answer: The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.

3. (a) The technical specialist was reassigned to the robotics lab. (b) His patent on pneumatic grippers was approved.

Answer: The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.

4. (a) This microcontroller features ultra-low power sleep states. (b) It is ideal for battery-operated remote sensors.

Answer: This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.

5. (a) The field researchers surveyed the wind turbine farm. (b) They submitted their acoustic emission assessment yesterday.

Answer: The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1 – Real Conditionals) Complete the sentences with the correct form of the verbs in brackets:

1. If the bus voltage _____ (exceed) 24V, the zener diode will divert the surge to ground.

Answer: exceeds

2. The watchdog timer will reset the CPU if the main thread _____ (fail) to ping it within 500 ms.

Answer: fails

3. If the operator _____ (engage) the safety latch, the hydraulic press will not actuate.

Answer: engages

4. If temperature levels _____ (rise) above 90°C, the cooling fans will switch to maximum RPM.

Answer: rise

5. The system will trigger an auditory alarm if the toxic gas detector _____ (register) more than 20 ppm.

Answer: registers

Exercise 8: If-Clauses (Type 3 – Unreal Past / Troubleshooting) Fill in the blanks with the past unreal conditional form (had + past participle / would have + past participle):

1. If the backup generator _____ (start) immediately, the data centre servers would not have shut down.

Answer: had started

2. The turbine bearing would not have seized if the maintenance crew _____ (lubricate) it as scheduled.

Answer: had lubricated

3. If the security team had patched the zero-day exploit earlier, the data breach _____ (prevent).

Answer: would have been prevented

4. The circuit breaker _____ (trip) safely if the overcurrent sensor had functioned properly.

Answer: would have tripped

5. If the telemetry link _____ (remain) stable, ground engineers would have corrected the satellite's orbital trajectory.

Answer: had remained

Exercise 9: Infinitives of Purpose

Combine the two ideas using an infinitive of purpose (to + verb):

1. (a) The mechanics installed dual damper springs. (b) They wanted to isolate chassis resonance.

Answer: The mechanics installed dual damper springs to isolate chassis resonance.

2. (a) Technicians purge the fuel line with dry nitrogen gas. (b) They want to remove residual flammable vapors.

Answer: Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.

3. (a) We introduced an optical encoder. (b) We intended to determine spindle velocity accurately.

Answer: We introduced an optical encoder to determine spindle velocity accurately.

4. (a) The developers wrote an automated load testing script. (b) Their goal was to simulate ten thousand concurrent connections.

Answer: The developers wrote an automated load testing script to simulate ten thousand concurrent connections.

5. (a) Gold plating is applied to terminal connectors. (b) Its purpose is to prevent atmospheric corrosion.

Answer: Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

Fill in the blanks with the gerund form of the bracketed verb after the preposition:

1. The aerodynamic team improved aircraft fuel economy by _____ (redesign) the winglet profile.

Answer: redesigning

2. Always disconnect the power supply before _____ (replace) defective surface-mount components.

Answer: replacing

3. The facility was praised for _____ (reduce) volatile organic chemical emissions by 40%.

Answer: reducing

4. Avoid _____ (clean) precision optical lenses with abrasive industrial solvents.

Answer: cleaning

5. The digital signal processor isolates target audio signals by _____ (filter) out background ambient noise.

Answer: filtering

Exercise 11: Impersonal Passive Voice

Convert the following active laboratory/industrial actions into the Impersonal Passive Voice:

1. The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Answer: 50 mL of hydrochloric acid was poured into the titration flask.

2. The automation unit tested the robotic arm under a 50 kg dynamic payload.

Answer: The robotic arm was tested under a 50 kg dynamic payload.

3. Field engineers calibrated the optical pyrometer prior to ignition.

Answer: The optical pyrometer was calibrated prior to ignition.

4. The technicians observed severe cavitation damage along the impeller blades.

Answer: Severe cavitation damage was observed along the impeller blades.

5. We store volatile solvents in flame-resistant containment cabinets.

Answer: Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

Fill in the blanks with the appropriate adverbial form derived from the adjective in brackets:

1. The stepper motor rotates _____ (precise) in 1.8-degree angular increments.

Answer: precisely

2. The telemetry ground station _____ (periodic) polls all deployed marine buoy sensors.

Answer: periodically

3. The thermal imaging camera registered the temperature drop _____ (rapid) during quenching.

Answer: rapidly

4. Structural joints must be welded _____ (firm) to withstand heavy cyclic vibrational stress.

Answer: firmly

5. The autonomous drone traversed the confined pipeline corridor _____ (smooth).

Answer: smoothly

Exercise 13: Reported Speech (Interrogative Structures) Transform each direct technical query into indirect/reported speech:

1. The quality manager asked the technician, "Has the insulation resistance test been documented?"

Answer: The quality manager asked the technician whether the insulation resistance test had been documented.

2. The lead architect enquired, "Can the server cluster handle double the transaction throughput?"

Answer: The lead architect enquired whether the server cluster could handle double the transaction throughput.

3. The field auditor asked the safety engineer, "Did you inspect the gas leakage relief lines?"

Answer: The field auditor asked the safety engineer whether he/she had inspected the gas leakage relief lines.

4. The client asked the team, "Will the firmware update fix the intermittent battery drain?"

Answer: The client asked the team whether the firmware update would fix the intermittent battery drain.

5. The instructor asked the lab group, "Why did the multimeter show fluctuating voltage readings?"

Answer: The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns Identify the incorrect noun form and rewrite each sentence correctly:

1. The procurement officer purchased several new machineries and equipments for the workshop.

Answer: Incorrect: machineries, equipments.

Correct: The procurement officer purchased several new machines and some equipment for the workshop.

2. The laboratory generated a huge amount of test datas during the wind tunnel trials.

Answer: Incorrect: datas.

Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.

3. We need three more softwares licenses to complete the finite element analysis on the model.

Answer: Incorrect: softwares.

Correct: We need three more software licenses to complete the finite element analysis on the model.

4. The field report provided numerous valuable informations regarding the soil composition.

Answer: Incorrect: informations.

Correct: The field report provided a lot of valuable information regarding the soil composition.

5. All chemical apparatuses and glasswares must be washed thoroughly after the experiment.

Answer: Incorrect: glasswares (and apparatuses).

Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms) Fill in the blanks using the correct irregular comparative form (better, worse, farther / further, less, more):

1. Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.

Answer: worse

2. The redesigned microstrip antenna achieved a _____ (good) signal-to-noise ratio than the baseline patch.

Answer: better

3. The autonomous vehicle traveled 50 km _____ (far) on a single charge after the weight reduction overhaul.

Answer: farther

4. Switching to asynchronous task queues resulted in _____ (little) CPU idle time during peak batch processing.

Answer: less

5. We require _____ (further / father) metallurgical investigation into why the stress cracks formed along the weld seam.

Answer: further

PART - B

Answer ALL questions.

Exercise 1: Tenses (Past Tense Conversion & Industrial Reporting) Rewrite the following active operational and lab descriptions into the Simple Past Tense:

1. "The technician inspects the solar arrays and records the peak power conversion value."

Answer: The technician inspected the solar arrays and recorded the peak power conversion value.

2. "The chemical analyzer detects trace mercury levels and sends an automated alert to the environmental dashboard."

Answer: The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.

3. "Maintenance crews lubricate the heavy conveyor bearings and replace the worn rubber belts before the morning shift."

Answer: Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. "The supervisory software polls field sensors every five milliseconds and archives temperature trends into a secure database."

Answer: The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.

5. "The testing team applies a 500-volt dielectric pulse across the stator winding and monitors for insulation breakdown."

Answer: The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Adjectives (Technical Metrics & Trade-Offs) Draft technical comparative sentences based on the prompts provided:

1. Write two comparative sentences contrasting Cloud Storage with On-Premise Physical Servers in terms of initial deployment cost and elastic scalability.

Answer: Cloud storage has a lower initial deployment cost than on-premise physical servers, because the company does not have to buy expensive hardware in advance. It is also much more scalable than physical servers, since extra capacity can be added within minutes instead of waiting for new machines to be installed.

2. Write two comparative sentences contrasting Fiber-Optic Cables with Copper Twisted-Pair Cables regarding data transfer bandwidth and susceptibility to electromagnetic interference.

Answer: Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables, so they can carry more data over longer distances. They are also far less susceptible to electromagnetic interference, whereas copper cables easily pick up noise from nearby power lines and motors.

3. Write two comparative sentences contrasting Electric Vehicles (EVs) with Internal Combustion Engine (ICE) Vehicles focusing on tailpipe emissions and initial purchase price.

Answer: Electric vehicles produce lower tailpipe emissions than ICE vehicles, since they release no exhaust gases while driving. However, EVs usually have a higher initial purchase price than comparable ICE vehicles because of their costly battery packs.

4. Write two comparative sentences comparing Hydraulic Actuators with Electromechanical Actuators in terms of raw force output and risk of fluid leakage.

Answer: Hydraulic actuators can generate a greater raw force output than electromechanical actuators of the same size, which makes them better for heavy presses and construction machines. On the other hand, they carry a higher risk of fluid leakage, while electromechanical actuators are cleaner and need less maintenance.

5. Write two comparative sentences comparing Relational (SQL) Databases with Document (NoSQL) Databases with respect to schema rigidity and horizontal query throughput.

Answer: Relational (SQL) databases have a more rigid schema than document (NoSQL) databases, because every table must follow a fixed structure, whereas documents can have flexible fields. Document databases generally achieve higher horizontal query throughput than SQL databases, since their data can be spread easily across many servers.

Exercise 3: Relative Clauses (Sentence Synthesis) Combine each pair of technical statements into a single cohesive sentence using a defining or non-defining relative clause (who, which, that, whose, or where):

1. (a) The algorithm analyzes microscopic cracks in turbine blades. (b) It was patented by the aeronautics team.

Answer: The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.

2. (a) The lead chemical engineer authorized the emergency boiler shutdown. (b) She noticed an abnormal pressure spike in chamber two.

Answer: The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.

3. (a) The laboratory houses a 400 kV impulse voltage generator. (b) High-voltage insulator puncture tests are conducted there.

Answer: The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.

4. (a) We engaged an external cybersecurity consultant. (b) His firmware vulnerability audit exposed three backdoor ports.

Answer: We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.

5. (a) The supervisory system utilizes an artificial neural network. (b) It predicts transformer insulation life based on dissolved gas concentrations.

Answer: The supervisory system utilizes an artificial neural network, which predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses (Technical Conditionals & Troubleshooting) Complete the following conditional statements logically and grammatically to depict realistic engineering scenarios:

1. (a) If the operating temperature exceeds 85°C, _____.

(b) If the operator had pressed the emergency shutdown switch immediately, _____.

Answer: (a) the cooling system will switch on automatically

(b) the motor would not have been damaged

2. (a) If the supply voltage drops below the 200V threshold, _____.

(b) If the network administrator had enabled multi-factor authentication, _____.

Answer: (a) the under-voltage relay will disconnect the load to protect the equipment

(b) the hackers would not have gained access to the server

3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, _____.

(b) If the mechanical team had checked the bearing lubrication schedule, _____.

Answer: (a) the ventilation fans will start and an alarm will sound

(b) the bearing would not have failed

4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, _____.

(b) If the thermal sensor had registered the runaway reaction in time, _____.

Answer: (a) it will stop immediately until the path is clear

(b) the operators would have stopped the process and prevented the explosion

5. (a) If the software receives an unauthenticated API token, _____.

(b) If the civil inspectors had reviewed the core sample density tests, _____.

Answer: (a) it will reject the request and return an "access denied" error

(b) they would have found the weak concrete before construction began

Exercise 5: Impersonal Passive Voice

Convert the following active sentences into the objective Impersonal Passive Voice suitable for formal engineering reports:

1. "The chemical engineers distilled the crude extract at a constant temperature of 60°C."

Answer: The crude extract was distilled at a constant temperature of 60°C.

2. "The metallurgist polished the cross-section of the weld seam with fine diamond paste."

Answer: The cross-section of the weld seam was polished with fine diamond paste.

3. "The quality control division rejected fifteen cast-iron pump casings due to internal porosities."

Answer: Fifteen cast-iron pump casings were rejected due to internal porosities.

4. "The technician measured the resistance across the bridge circuit using a calibrated digital ohmmeter."

Answer: The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.

5. "The flight test crew initiated the emergency aerodynamic braking flaps at an altitude of 3,000 meters."

Answer: The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech (Interrogative Technical Transformations) Transform the following direct engineering queries into Reported (Indirect) Speech:

1. The safety supervisor asked the intern, "Did you verify the grounding connections on the generator?"

Answer: The safety supervisor asked the intern whether he/she had verified the grounding connections on the generator.

2. The plant director asked the site engineer, "Has the emergency effluent treatment plant received clearance?"

Answer: The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.

3. The IT security lead enquired of the developer, "Why did you disable the firewall rule on the database server?"

Answer: The IT security lead enquired of the developer why he/she had disabled the firewall rule on the database server.

4. The client asked the project manager, "Can the bridge structure withstand Category 4 hurricane winds?"

Answer: The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.

5. The research guide asked the postgraduate scholar, "How many sample trials did you conduct under cryogenic conditions?"

Answer: The research guide asked the postgraduate scholar how many sample trials he/she had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:

1. (a) We need to install several new software and machineries in the CAD laboratory.

(b) The team collected plenty of useful informations from the field test.

Answer: (a) Errors: software (several), machineries. Correct: We need to install some new software and machinery in the CAD laboratory.

(b) Error: informations. Correct: The team collected plenty of useful information from the field test.

2. (a) The technician ordered ten new apparatuses and modern electronic equipments for the antenna lab.

(b) High-speed train operations require advanced transport infrastructures and dedicated electrical powers.

Answer: (a) Errors: apparatuses, equipments. Correct: The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.

(b) Errors: infrastructures, powers. Correct: High-speed train operations require advanced transport infrastructure and dedicated electrical power.

3. (a) The environmental sensor gathers extensive atmospheric datas and pollutions indices every minute.

(b) The construction crew unloaded tons of sands, cements, and steel debris at the worksite.

Answer: (a) Errors: datas, pollutions. Correct: The environmental sensor gathers extensive atmospheric data and pollution indices every minute.

(b) Errors: sands, cements. Correct: The construction crew unloaded tons of sand, cement, and steel debris at the worksite.

4. (a) All electrical hardwares and wirings must comply with local safety legislations.

(b) The data scientist discovered several interesting knowledges after running the machine learning clustering.

Answer: (a) Errors: hardwares, wirings, legislations. Correct: All electrical hardware and wiring must comply with local safety legislation.

(b) Error: knowledges. Correct: The data scientist discovered several interesting insights after running the machine learning clustering.

5. (a) Laboratory staffs must clear away all glasswares and acidic wastes before leaving the premises.

(b) The project failed due to a lack of proper fundings and technical researches.

Answer: (a) Errors: staffs, glasswares, wastes. Correct: Laboratory staff must clear away all glassware and acidic waste before leaving the premises.

(b) Errors: fundings, researches. Correct: The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose & Gerund Constructions Rewrite the following pairs of sentences using either an Infinitive of Purpose (to + verb) or a Gerund construction after a preposition (by / for + -ing):

1. (a) A pressure relief valve is installed. It vents excess boiler steam.

(b) The automated script runs every midnight. It archives legacy server logs.

Answer: (a) A pressure relief valve is installed to vent excess boiler steam.

(b) The automated script runs every midnight to archive legacy server logs.

2. (a) Automotive engineers installed an intercooler. It reduces the compressed air intake temperature.

(b) The technicians analyzed the structural stress points. They placed piezoelectric strain gauges along the girder.

Answer: (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.

(b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.

3. (a) High-performance heatsinks feature copper heatpipes. They conduct heat away from the silicon die.

(b) The factory cut water consumption by 30%. They implemented a closed-loop recycling filtration system.

Answer: (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.

(b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.

4. (a) Safety guards are mounted over rotating flywheel belts. They prevent accidental worker contact.

(b) The algorithm optimizes memory bandwidth. It pre-fetches sequential array indices into L2 cache.

Answer: (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.

(b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.

5. (a) The electronics lab installed an uninterruptible power supply (UPS). It protects servers against sudden voltage sags.

(b) The metallurgy group enhanced surface hardness. They diffused nitrogen atoms into the steel casing.

Answer: (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.

(b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner & Operational Precision Fill in the blanks with the appropriate adverbial forms derived from the words provided in brackets:

1. (a) The drone adjusted its flight path _____ (automatic) to avoid overhead power cables.

(b) The software patch was integrated _____ (flawless) into the production server.

Answer: (a) automatically

(b) flawlessly

2. (a) The CNC cutting head traced the complex automotive profile _____ (accurate).

(b) The turbine governor responded _____ (instant) to the unexpected grid frequency drop.

Answer: (a) accurately

(b) instantly

3. (a) The hazardous gas extraction vents operate _____ (continuous) throughout the electroplating process.

(b) Technicians handled the radioactive isotope capsule _____ (cautious) using robotic arm manipulators.

Answer: (a) continuously

(b) cautiously

4. (a) The optical pyrometer recorded the furnace temperature _____ (consistent) across three production shifts.

(b) The structural frame buckled _____ (catastrophic) under the simulated earthquake test load.

Answer: (a) consistently

(b) catastrophically

5. (a) The firmware routine checks for incoming serial packets _____ (periodic) every hundred microseconds.

(b) The titanium alloy workpiece was machined _____ (smooth) to a mirror finish.

Answer: (a) periodically

(b) smoothly

Exercise 10: Modals of Capability, Obligation, and Prohibition Complete the sentences by inserting an appropriate modal auxiliary verb that conveys the specific operational meaning specified in brackets:

1. (a) The microcontroller _____ (capability) drive up to six stepper motors simultaneously.

(b) Laboratory staff _____ (strict prohibition) handle corrosive reagents without full face shields.

Answer: (a) can

(b) must not

2. (a) Before entering the cleanroom chamber, all personnel _____ (mandatory obligation) put on antistatic footwear.

(b) Given the high ambient humidity, water droplet condensation _____ (moderate possibility) form inside the optical housing.

Answer: (a) must

(b) may

3. (a) Unauthorized maintenance personnel _____ (strict prohibition) override the high-voltage interlock switch.

(b) The upgraded wireless transmitter _____ (past capability) broadcast telemetry signals up to forty kilometers.

Answer: (a) must not

(b) could

4. (a) Quality inspection teams _____ (strong recommendation / advice) recalibrate measuring calipers every ninety days.

(b) The digital ammeter is reading zero amperes; the auxiliary fuse _____ (logical deduction / certainty) be blown.

Answer: (a) should

(b) must

5. (a) Industrial robots _____ (capability) perform repetitive spot welds with sub-millimeter precision.

(b) Visitors _____ (strict prohibition) enter the laser testing enclosure while the amber warning beacon is active.

Answer: (a) can

(b) must not

PART - C

Answer ALL questions.

Topic 1: Writing Imperative Safety Instructions (CO1 | BL3) Target: 8 clear, numbered, imperative instructions using modals/action verbs.

1. Question 1 : Write a set of eight clear, imperative safety instructions to be strictly displayed and followed inside a Chemistry & Materials Testing Laboratory to avoid chemical burns, inhalation hazards, and equipment contamination.

Answer:

SAFETY INSTRUCTIONS – CHEMISTRY & MATERIALS TESTING LABORATORY

1. Wear a lab coat, safety goggles, nitrile gloves and closed shoes at all times inside the laboratory.

2. Handle all acids, alkalis and volatile solvents inside the fume hood, and never smell or inhale chemicals directly.

3. Always add acid to water slowly; never add water to acid, to avoid splashing and chemical burns.

4. Label every bottle and beaker clearly, and return all reagents to their proper shelves after use.

5. Use separate clean spatulas, pipettes and glassware for each chemical to prevent contamination of samples and equipment.

6. Clean spills immediately with the spill kit and inform the instructor at once; flush affected skin or eyes with water for at least 15 minutes.

7. Do not eat, drink or store food in the laboratory, and wash your hands thoroughly before leaving.

8. Dispose of chemical waste only in the marked waste containers, and switch off all burners, gas lines and equipment before you leave.

2. Question 2 : Write a set of eight clear, imperative safety instructions to be strictly displayed and observed inside a High-Voltage Electrical Machines & Drives Laboratory to prevent electric shock, arc flash hazards, and accidental short circuits during motor loading experiments.

Answer:

SAFETY INSTRUCTIONS – HIGH-VOLTAGE ELECTRICAL MACHINES & DRIVES LABORATORY

1. Wear insulated footwear, stand on the rubber mat, and remove rings, watches and metal chains before starting any experiment.

2. Check every connection carefully and get the instructor's approval before you switch on the supply.

3. Never touch live terminals, bare conductors or rotating shafts; switch off and isolate the supply before changing any connection.

4. Use only insulated tools and undamaged cables of the correct rating, and make sure that fuses and circuit breakers match the machine rating.

5. Set the load rheostat or brake to the minimum position before you start the motor, and increase the load slowly and gradually.

6. Keep clear of the rotating parts, and tie back long hair and loose clothing to avoid entanglement.

7. Keep the floor and your hands dry, avoid working alone, and use only one hand to operate the switches whenever possible.

8. Learn the location of the emergency stop button, the CO_2 fire extinguisher and the first-aid box, and report every accident or fault to the instructor immediately.

3. Question 3: Write a set of eight clear, imperative safety guidelines to be posted in an Advanced Manufacturing and CNC Machining Workshop to prevent injuries from high-speed spindle operations, hot metallic chip expulsion, coolant spills, and moving cutter heads.

Answer:

SAFETY GUIDELINES – ADVANCED MANUFACTURING & CNC MACHINING WORKSHOP

1. Wear safety glasses, safety shoes and close-fitting clothes; tie back long hair and remove rings, watches and loose jewellery.

2. Do not wear gloves near a rotating spindle or cutter, because they can get caught and pull your hand in.

3. Close the safety door and check that all guards and interlocks are in place before you start the machine; never bypass them.

4. Clamp the workpiece and tighten the cutting tool firmly, and remove the chuck key or wrench before you start the spindle.

5. Keep your hands away from the spindle and cutter head, and stop the machine completely before you measure, clean or change a tool.

6. Clear hot metal chips only with a brush or chip hook once the machine has stopped; never use your bare hands or compressed air.

7. Wipe up coolant and oil spills immediately and keep the floor dry and free of chips to prevent slips and falls.

8. Run the program in dry-run or single-block mode first, keep your hand near the emergency stop button, and never leave a running machine unattended.

Topic 2: Free Writing – Festivals & Cultural Celebrations (CO2 | BL4) Target: Descriptive essay of 250 words integrating community customs, festive meals, and sustainable practices.

1. Question 1 : Write a descriptive essay of 250 words about "A Traditional Harvest Festival in Your Region and How Modern Eco-Friendly Practices Can Preserve Its Core Spirit." Discuss community customs, traditional festive meals, and sustainable celebration methods.

Answer:

Pongal: A Harvest Festival and Its Green Future

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated every January to thank the Sun God, the soil and the farm animals for a good harvest. For four days, the whole community comes together, and the festival shows how closely our culture is connected with agriculture.

The celebration begins with Bhogi, when families clean their houses and throw away old things. On Thai Pongal, the main day, fresh rice is cooked with milk and jaggery in a new clay pot in the open courtyard. When the milk boils over, everyone shouts "Pongalo Pongal!" On Mattu Pongal, farmers decorate and worship their cattle, and on Kaanum Pongal families visit relatives. Neighbours draw colourful kolams in front of their houses, and villages organise games and cultural programmes.

The festive meal is served on a banana leaf. It includes sweet pongal, venn pongal, sambar, vadai, sugarcane and a mixed vegetable kootu prepared with seasonal vegetables. Modern eco-friendly practices can protect the spirit of this festival. Instead of burning old tyres and plastic during Bhogi, which creates heavy smoke, people can compost dry waste and recycle old items. Clay pots, banana leaves and areca- leaf plates should replace plastic and thermocol. Kolams can be drawn with natural rice flour instead of chemical colours. Communities can use cloth bags, give leftover food to the needy or compost it, and light the evening programmes with solar lamps. By following these simple steps, we can enjoy Pongal happily and still keep our environment clean.

2. Question 2 : Write a descriptive essay of 250 words on "A Traditional Temple Car or Float Festival in Your Hometown and Sustainable Measures to Minimize Single-Use Plastics and Noise Pollution." Detail communal rituals, traditional food offerings, and eco-friendly crowd management methods.

Answer:

The Temple Car Festival of My Hometown

The temple car festival, known as the Ther Thiruvizha, is the biggest annual event in my hometown. Every year in the Tamil month of Panguni, thousands of people gather in front of the old temple to watch a huge wooden chariot, decorated with flowers, silk cloth and palm leaves, being pulled through the four main streets. The festival begins with the flag hoisting and continues for ten days with special poojas, abhishekams and processions of the deity on different vahanas.

On the day of the car festival, devotees take a holy bath, observe a fast in the morning and join together to pull the thick ropes of the chariot while chanting "Hara Hara". People of all communities and ages work as volunteers. Many families offer food as prasadam, such as pongal, sundal, curd rice, tamarind rice and sweet payasam. Free meals called annadhanam are served to hundreds of people on banana leaves.

However, the festival also creates plastic waste, loud noise from speakers and crackers, and heavy crowding. To reduce these problems, the temple committee can ban single-use plastic cups, bags and bottles, and provide steel tumblers, areca-leaf plates and drinking water stations. Separate bins for wet and dry waste should be placed, and leftover food and flowers can be composted. Loudspeakers must be kept within the permitted decibel limit, and firecrackers should be avoided. For crowd management, barricades, one-way routes, clear signboards, trained volunteers, first-aid camps and mobile toilets must be arranged. With these measures, the ancient festival can continue peacefully and cleanly.

3. Question 3 : Write a descriptive essay of 250 words on "Celebrating Regional Folk Art and Handicraft Fairs Using Renewable Solar Micro-Grids and Zero-Waste Composting." Discuss the cultural atmosphere, heritage delicacies, and modern environmental protection strategies.

Answer:

Folk Art and Handicraft Fairs: Tradition Powered by the Sun

Regional folk art and handicraft fairs are a colourful celebration of our heritage. They bring artisans from many villages under one roof and give visitors a chance to see the skills that have been passed down for generations.

The atmosphere at such a fair is full of life. The beat of the thappu and parai drums, and the graceful steps of karagattam, oyilattam and puppet shows, attract large crowds. Stalls display Thanjavur paintings, wooden toys, terracotta pots, palm-leaf baskets, handloom sarees and brass lamps, and visitors can watch the artists at work. The food stalls serve heritage delicacies such as kambu koozh, ragi kali, adhirasam, murukku, kuzhi paniyaram and filter coffee, which are made with local grains and traditional methods.

To protect the environment, modern technology can be added without losing the charm of the fair. A solar micro-grid, with rooftop panels and a battery bank, can supply clean electricity to the lights, sound systems and mobile charging points, so noisy and polluting diesel generators are not needed. A zero- waste plan is equally important. Food waste and leaves can be collected in separate bins and turned into compost for local farms, and cloth bags, steel tumblers and leaf plates can replace single-use plastic. Volunteers can guide visitors to sort the waste correctly, and the money saved on fuel and waste disposal can be used to support the artisans. In this way, the fair keeps the cultural spirit alive and also teaches the young generation to protect nature.

Topic 3: Free Writing – Food & Technical Travel Plans (CO2 | BL4) Target: Structured travel narrative (250 words) combining route logistics, local dining experiences, and technical landscape observations.

1. Question 1 : Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to a Major Hydroelectric Dam." Include route details, logistics, local cultural food experiences, and observations of the plant's operational landscape.

Answer:

A Technical Study Tour to the Mettur Dam Route and logistics: Our group of forty first-year students will leave the college by a hired bus at 5.00 a.m. and travel from Chennai to Mettur in Salem district through the Chennai–Salem highway. The journey is about 330 km and takes nearly seven hours with a tea break at Villupuram. We will carry our identity cards, a notebook, a water bottle and a safety helmet, and the visit permission will be taken from the electricity board in advance.

Food experience: On the way, we will have a simple breakfast of idli, pongal and vadai in a roadside restaurant. For lunch, the guide has suggested a local mess near the dam where we can taste a traditional meal on a banana leaf, with fresh Cauvery river fish fry, ragi kali and kambu koozh, which are popular in the region.

Technical observations: The Mettur Dam, built across the River Cauvery, is one of the largest dams in India. From the top, we can see the huge reservoir, the long masonry wall, the spillway gates and the surge pool. In the power house, engineers will explain how water from the reservoir flows through the penstock to the turbine blades, which rotate the generator and produce electricity. We will observe the control room, the transformer yard and the transmission lines that carry power to the grid. The guide will also tell us how the dam supports irrigation, drinking water supply and flood control.

We will return in the evening after a short visit to the dam park, with a better understanding of renewable energy.

2. Question 2 : Write a structured travel plan and descriptive feature (250 words) on "A Technical Study Tour to an Integrated Solar and Wind Hybrid Energy Farm." Detail route planning, transit logistics, local culinary specialties sampled en route, and operational observations of the wind turbine park and photovoltaic arrays.

Answer:

A Study Tour to a Solar–Wind Hybrid Energy Farm Route planning: Our department has planned a one-day study tour to a hybrid energy farm in the wind-rich region of southern Tamil Nadu, near Muppandal. The college bus will start from Chennai at night, and the group of fifty students will travel by road via Trichy and Madurai to Tirunelveli and Nagercoil, a distance of about 700 km. We will begin the journey the evening before and reach the farm the next morning. Each student will carry a cap, sunglasses, a helmet and a water bottle.

Food experience: On the way we will have breakfast at a local hotel in Tirunelveli and taste the famous Tirunelveli halwa and hot idli with sambar. In the Nagercoil region, we will try banana chips, kari dosa and a simple fish curry meal, which are specialities of the area.

Operational observations: The farm is a perfect example of hybrid energy. Tall wind turbines with three blades rotate when the strong wind blows, and the nacelle on each tower houses the gearbox and generator. Beside them, rows of photovoltaic panels convert sunlight into electricity. Engineers will explain that the two sources balance each other, because wind is stronger during the monsoon season and the sun shines stronger in summer. We will also see the inverters, the substation and the control room where the power output is monitored on large screens, and we will learn how the panels are cleaned and maintained. The tour will end with a feedback session, and each student will submit a short report on renewable energy.

3. Question 3 : Write a structured travel plan and descriptive feature (250 words) on "A Technical Field Visit to an Automated Seawater Reverse Osmosis (RO) Desalination Plant."

Describe transportation details, regional coastal food experiences, and observations of intake pumping stations, membrane filtration racks, and brine discharge outlets.

Answer:

A Field Visit to a Seawater Reverse Osmosis Desalination Plant Transportation details: Our class of forty students will visit the seawater desalination plant at

Nemmeli on the East Coast Road. The college bus will leave the campus on OMR at 8.30 a.m., and the plant is only about 25 km away, so we will reach there in less than an hour. Students must carry their identity cards, closed shoes and a notebook, and we will return to the college by 4.00 p.m.

Coastal food experience: After the visit, we will have lunch in a small restaurant on the coast near Mahabalipuram. It is famous for fresh seafood, so students who eat non-vegetarian food can try fish fry, prawn masala and crab curry with hot rice, while vegetarians can enjoy lemon rice, sundal and fresh coconut water.

Technical observations: At the plant, engineers will take us first to the intake pumping station, where seawater is drawn through pipes from the sea and pumped to the pre-treatment unit to remove sand and suspended particles. Next, we will see the long racks of membrane filtration, which hold hundreds of reverse osmosis membranes. High-pressure pumps push the seawater through them, and only fresh water passes through while the salt is left behind. The plant is highly automated, and the entire process is monitored from the control room through sensors and a SCADA system. Finally, we will observe the brine discharge outlet, where the concentrated salt water is carefully released far into the sea to reduce the harm to marine life. This visit will help us understand how technology solves water scarcity.

Topic 4: Formal Letters of Request (CO3 | BL4) Target: Formal business letter layout with sender/receiver address, clear subject line, technical rationale, specific requirements, and courteous sign-off.

1. Question 1 : Draft a formal Letter of Request to the Head of the Computer Science Department requesting permission and lab access after college hours for your project team to prepare for a National AI Hackathon. Outline project objectives and required software resources.

Answer:

From:

Harini G

First-Year B.E. Computer Science and Engineering St. Joseph's College of Engineering

OMR, Chennai – 600 119

Date: 01 November 2026

To:

The Head of the Department,

Department of Computer Science and Engineering, St. Joseph's College of Engineering,

OMR, Chennai – 600 119

Respected Sir/Madam,

Subject: Request for permission to use the Computer Lab after college hours for National AI

Hackathon preparation

I am writing on behalf of my four-member project team to request your kind permission to use the Computer Science laboratory after regular college hours. Our team has been shortlisted for the National AI Hackathon, which will be held next month, and we need extra time to prepare our project.

Project objective: Our project is an AI-based crop disease detection system that identifies leaf

diseases from photographs using a convolutional neural network. We plan to train the model, test its accuracy and build a simple mobile interface for farmers.

Resources required: We need systems with Python 3, TensorFlow/Keras, OpenCV, Jupyter

Notebook, Git and Visual Studio Code installed, along with a stable internet connection to download datasets from open sources.

Timing requested: We request access to the laboratory from 4.30 p.m. to 7.00 p.m. on weekdays for the next three weeks. We assure you that we will follow all the lab rules, take care of the equipment and switch off all systems before leaving. Our faculty guide has agreed to supervise us.

I kindly request you to grant us the permission so that we can represent our department with confidence.

Yours faithfully,

Harini G First-Year B.E. Computer Science and Engineering

2. Question 2 (Original - Industrial Visit): Write a formal Letter of Request to the Plant Manager of an Industrial Robotics & Automation Factory seeking approval for an industrial plant visit for 50 first-year engineering students, outlining the intended learning outcomes.

Answer:

From:

Harini G

First-Year B.E. Computer Science and Engineering St. Joseph's College of Engineering

OMR, Chennai – 600 119

Date: 01 November 2026

To:

The Plant Manager,

[Name of the Company]

Industrial Robotics & Automation Factory,

[Address]

Respected Sir/Madam,

Subject: Request for permission to conduct an industrial visit for first-year engineering students

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai, and I am writing on behalf of my class to request your kind permission to visit your robotics and automation factory. We are a group of 50 students accompanied by two faculty members. At present we are studying the basics of engineering, programming and electronics. We believe that seeing industrial robots and automated production lines in action will help us connect classroom learning with real applications.

Intended learning outcomes: (i) to understand how robotic arms, sensors, controllers and conveyors work together in a production line; (ii) to learn about industrial safety practices and quality control; (iii) to observe how programming and electronics are used in automation; and (iv) to get awareness of

career opportunities in robotics and automation. We request you to permit the visit on any convenient weekday in the next month, preferably between 10.00 a.m. and 1.00 p.m. We will follow all the safety rules and instructions of your company, and we will make our own transport arrangements. We would be grateful if one of your engineers could give a short briefing.

I hope you will consider our request favourably.

Yours faithfully,

Harini G

First-Year B.E. Computer Science and Engineering

3. Question 3 (Practice A - Workshop Facilities): Draft a formal Letter of Request to the Head of the Mechanical Engineering Department seeking permission and machine shop access after college hours for your student team to fabricate and test an energy-efficient electric go-kart for an inter- collegiate design challenge.

Answer:

From:

Harini G

First-Year B.E. Computer Science and Engineering St. Joseph's College of Engineering

OMR, Chennai – 600 119

Date: 01 November 2026

To:

The Head of the Department,

Department of Mechanical Engineering,

St. Joseph's College of Engineering,

OMR, Chennai – 600 119

Respected Sir/Madam,

Subject: Request for machine shop access after college hours to build an electric go-kart

We are a team of six students who have registered for the Inter-Collegiate Design Challenge, which will be held next month. We would like to design, fabricate and test an energy-efficient electric go-kart, and we respectfully request your permission to use the machine shop after college hours.

Project objective: Our aim is to build a lightweight go-kart powered by a brushless DC motor and a lithium-ion battery pack, with a target speed of 40 km/h and a good driving range on a single charge.

Facilities required: We need access to the welding unit, lathe, drilling machine, grinding machine and cutting tools to fabricate the chassis and mount the motor, as well as an open space to test the vehicle.

Timing requested: We request permission to use the workshop from 4.30 p.m. to 7.00 p.m. on weekdays and on Saturdays until 1.00 p.m., for the next four weeks. We will work only under the guidance of the lab technician and our faculty advisor, wear safety equipment and keep the workshop clean.

We hope that you will kindly approve our request and support our team in representing the college.

Yours faithfully,

Harini G

First-Year B.E. Computer Science and Engineering

4. Question 4 (Practice B - Sponsorship & Duty Leave): Draft a formal Letter of Request to the College Principal / Dean of Student Affairs requesting financial sponsorship and duty leave for a five-member multidisciplinary student team selected to present a drone surveillance prototype at a National Defense Innovation Expo.

Answer:

From:

Harini G

First-Year B.E. Computer Science and Engineering St. Joseph's College of Engineering

OMR, Chennai – 600 119

Date: 01 November 2026

To:

The Principal,

St. Joseph's College of Engineering,

OMR, Chennai – 600 119

(Through the Dean of Student Affairs)

Respected Sir/Madam,

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo I am pleased to inform you that our five-member multidisciplinary team, with students from Computer Science, Electronics and Mechanical Engineering, has been selected to present its drone surveillance prototype at the National Defense Innovation Expo, to be held from 12 to 14 December 2026. This is a great opportunity to represent our college at the national level.

About the prototype: Our drone is fitted with a camera, GPS and obstacle-avoidance sensors, and it uses image processing to detect unauthorised movement in restricted areas.

Sponsorship requested: We request financial support of about ₹30,000 to cover travel and accommodation for five students (₹18,000), exhibit setup and transportation of the prototype (₹7,000), and registration and printing charges (₹5,000).

Duty leave: We also request duty leave for all five team members from 11 to 15 December

2026, including travel days. We will complete the missed assignments and attend the extra classes arranged by our faculty after returning.

We assure you that we will present our college with discipline and pride, and we will submit a detailed report of the expo. Your kind support will be of great encouragement to us. Yours faithfully,

Harini G

First-Year B.E. Computer Science and Engineering

Topic 5: Compare & Contrast Descriptive Essays (CO4 | BL5) Target: Analytical essay (250–300 words) using comparative transitions (whereas, conversely, in contrast, on the other hand) to evaluate technical trade-offs.

1. Question 1 : Write a comprehensive Compare and Contrast essay (250–300 words) analyzing the Positives and Negatives of Electric Vehicles (EVs) versus Conventional Internal Combustion

Engine (ICE) Vehicles. Address battery life, manufacturing footprint, operating costs, and charging network feasibility.

Answer:

Electric Vehicles versus Internal Combustion Engine Vehicles

Electric vehicles (EVs) and conventional internal combustion engine (ICE) vehicles are two very different ways of travelling, and each has its own strengths and weaknesses. Regarding battery life, an EV battery usually lasts eight to ten years, but it slowly loses capacity and is expensive to replace. In contrast, an ICE engine can run for many years with regular servicing, although it has many moving parts that need oil changes and repairs. Considering the manufacturing footprint, EVs produce more carbon during production because of the mining of lithium and cobalt, whereas ICE vehicles have a smaller production footprint. However, over its lifetime an EV releases much less pollution because it has no tailpipe emissions, while an ICE vehicle keeps releasing carbon dioxide and other harmful gases. When it comes to operating costs, EVs are cheaper to run, since electricity costs less than petrol and there are fewer parts to repair. On the other hand, ICE vehicles need regular fuel and maintenance, which increases the long-term expense, though their purchase price is generally lower.

The charging network is the biggest drawback of EVs. Charging stations are still limited, especially in small towns, and charging takes much longer than refuelling. Conversely, petrol bunks are available almost everywhere, and a tank can be filled within a few minutes. In conclusion, EVs are cleaner and cheaper to operate, whereas ICE vehicles are more convenient at present. As charging infrastructure and battery recycling improve, EVs are likely to become the better choice for the future.

2. Question 2 : Compose an analytical essay comparing the Positives and Negatives of Remote / Virtual Internships versus On-Site Physical Internships for early-stage engineering students, evaluating hands-on skill acquisition, schedule flexibility, mentorship, and professional networking.

Answer:

Remote Internships versus On-Site Internships

For engineering students, an internship is the first step from classroom learning to real industry work. Today, students can choose between remote (virtual) internships and on-site (physical) internships, and each has clear advantages and disadvantages.

In terms of hands-on skill acquisition, on-site internships are better, because students can use real machines, laboratories and equipment, and they learn by watching and working with professionals. In contrast, remote internships are mostly limited to software, simulation and documentation tasks, so students studying mechanical or electrical branches may not get practical experience. However, remote internships still allow students to learn coding, data analysis and design tools very effectively.

Regarding schedule flexibility, remote internships are far superior. Students can work from home, save travel time and cost, and manage their college classes, exams and assignments easily. On the other hand, on-site internships have fixed working hours and require daily travel or even relocation, which can be tiring and expensive.

Mentorship is stronger in on-site internships, since a student can ask questions immediately and receive feedback in person. Remote mentors, on the other hand, are often busy and available only through scheduled online meetings, so doubts may remain unsolved for longer. Similarly, professional networking is easier in a physical office, where students meet engineers, managers and other interns daily. Virtual internships offer fewer personal interactions, although platforms like LinkedIn can partly make up for this. In conclusion, on-site internships are more suitable for practical skills and mentoring, whereas remote internships suit flexibility. A combination of both is the best choice.

3. Question 3 : Write an analytical Compare and Contrast essay (250–300 words) evaluating the Positives and Negatives of Cloud Computing Infrastructure versus On-Premise Physical Data Centers for engineering enterprises. Address upfront capital expenditure, data security compliance, recurring overheads, and workload scalability.

Answer:

Cloud Computing versus On-Premise Data Centers

Engineering companies today need powerful computing resources for design, simulation and data storage. They can either rent these resources from a cloud provider or build their own on-premise data centre, and the two options differ in several ways.

Regarding upfront capital expenditure, cloud computing is clearly cheaper to begin with, because the company pays only for what it uses and does not have to buy servers, cooling systems and buildings. In contrast, an on-premise data centre requires a very large initial investment in hardware, space and power backup.

When it comes to data security compliance, on-premise centres give the company full control over its sensitive designs and customer data, so it is easier to follow strict laws and internal policies. On the other hand, cloud data is stored on shared infrastructure, and the company depends on the provider's security. Although big providers offer strong encryption and certifications, concerns about data location and unauthorised access remain. Considering recurring overheads, an on-premise data centre has continuous costs such as electricity, cooling, maintenance and skilled staff. Cloud services, whereas, charge monthly usage fees, which are easy to predict for small workloads but can become very expensive for heavy and continuous use.

The biggest advantage of the cloud is workload scalability. Resources can be increased within minutes during heavy simulations and reduced afterwards. Conversely, expanding an on-premise centre needs new hardware, which takes time and money, and unused machines are wasted during low demand.

In conclusion, the cloud is best for flexibility and low initial cost, while on-premise centres suit enterprises needing strict control. Many companies now use a hybrid model to get the benefits of both.

4. Question 4 : Write a balanced Compare and Contrast essay (250–300 words) discussing the Positives and Negatives of 3D Printing (Additive Manufacturing) versus Traditional CNC Milling (Subtractive Manufacturing) in prototype fabrication. Evaluate material wastage, production speed for batch runs, surface finish accuracy, and geometric complexity limits.

Answer:

3D Printing versus CNC Milling in Prototype Fabrication

Prototypes help engineers to test an idea before mass production. Two popular methods for making them are 3D printing (additive manufacturing) and CNC milling (subtractive manufacturing), and they differ in many important ways.

The most visible difference is material wastage. In 3D printing, material is added layer by layer, so very little is wasted. In contrast, CNC milling cuts away a solid block, and a large amount of material becomes chips and scrap, although the metal chips can be recycled. Regarding production speed for batch runs, CNC milling is faster and more consistent once the program is ready, because the machine can produce many identical parts quickly. On the other hand, 3D printing is slow, since each part may take several hours to print, so it is better for one-off prototypes or small batches.

When we compare surface finish and accuracy, CNC milling gives smoother surfaces and tighter tolerances, which makes it suitable for parts like gears and shafts. Parts made by 3D printing often

show visible layer lines and need sanding or polishing, whereas high-end printers can improve this but at a higher cost.

However, 3D printing has a great advantage in geometric complexity. It can produce hollow structures, internal channels and lattice shapes in a single piece. Conversely, CNC milling is limited by tool access, so very complex internal shapes are difficult or impossible to cut. In conclusion, 3D printing is ideal for complex, low-volume prototypes with minimum waste, whereas CNC milling is better for accurate, smooth and repeated parts. The best choice depends on the design, the material and the budget of the project.

Topic 6: Personal Email Writing (CO5 | BL6) Target: Personal email (informal yet polite and articulate tone) conveying reflections, academic transitions, and structured study plans.

1. Question 1 : Draft a Personal Email to your high school mentor or teacher sharing your first-semester experiences at engineering college. Highlight the transition to independent campus life, your initial laboratory projects, and how you are preparing for your upcoming end-semester examinations.

Answer:

To: mentor@email.com

From: harini@email.com

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I am writing to share my experiences from my first semester at St. Joseph's College of Engineering. I still remember your advice before I left school, and it has helped me a lot.

At first, living independently on campus was not easy. I had to manage my time, my travel and my food on my own, and I missed home. But slowly I made new friends, joined a study group and learned to plan my day. I have realised how important discipline is. In the laboratories, we have started with simple projects. In the programming lab, I wrote my first C programs, and in the electronics and IoT lab, I connected an LED and a sensor to an Arduino board. When the circuit worked for the first time, I felt very proud. Now the end-semester examinations are coming. I have made a timetable, and I revise one subject every day. I solve previous question papers, practise programs in the lab, and prepare short notes for the final days. I also try to sleep well and avoid last-minute stress. I would be very happy to get your suggestions for scoring well. Please convey my regards to all the teachers.

With regards,

Harini G

2. Question 2 : Draft a personal email to your high school science teacher reflecting on your first term in engineering. Focus on adapting to hands-on engineering lab coursework, participating in your first technical club or hackathon, and your strategy for the upcoming university theory and practical exams.

Answer:

To: scienceteacher@email.com

From: harini@email.com

Subject: Reflections on my first term in engineering

Dear Madam/Sir,

Greetings! I hope you and everyone at school are well. I wanted to write to you because your science classes inspired me to choose engineering, and I would like to share how my first term has been.

The biggest change was the hands-on lab coursework. In school we did experiments in a small group, but here we build and test circuits ourselves and write detailed lab records. At first, I made many mistakes and my circuits did not work. With practice and the help of my lab instructor, I have learnt to debug step by step, and now I enjoy it.

I also joined the coding and robotics club of our college, and I took part in my first hackathon with three friends. We built a small sensor-based project in twenty-four hours. We did not win a prize, but we learnt about teamwork, time management and presentation skills. For the upcoming university theory and practical exams, my plan is simple. I study the theory subjects with a weekly schedule and revise the formulas daily. I solve model papers and question banks, and I practise all the lab experiments again so that I can explain the procedure clearly in the viva.

Your guidance still motivates me. Please give me your blessings and advice. Yours sincerely,

Harini G

3. Question 3 : Draft a personal email to your elder sibling or college alumnus seeking advice on balancing academic requirements with co-curricular technical clubs. Describe your daily schedule, challenges in managing lab record submissions, and ask for their tips on semester exam revision.

Answer:

To: elderbrother@email.com

From: harini@email.com

Subject: Need your advice on balancing studies and clubs

Dear Anna/Akka,

How are you? I hope your work is going well. I need your advice, because I am finding it difficult to balance my studies and my technical club activities.

My day starts at 6.00 a.m. I attend classes from 9.00 a.m. to 4.30 p.m., and after that I go to the technical club meeting for one hour. I reach home by 7.00 p.m., have dinner and study till about 10.30 p.m. On weekends, I work on club projects and assignments. My main problem is the lab record submissions. Each week I must complete the observation, the diagrams and the write-up for several experiments, and sometimes I leave them for the last day and get stressed. Because of the club work, I also lose time that I need for revising class topics.

As you have already completed the same course, could you please tell me how you managed everything? I would especially like your tips on:

1. preparing a weekly plan and finishing records on time,

2. choosing which club activities to join, and

3. revising for semester exams, including how many days to start before the exams and how to use previous question papers.

Your advice will be very valuable to me. Please reply when you are free, and take care of your health.

With love,

Harini G

Topic 7: Travel Planning & Budgetary Itineraries (CO5 | BL6) Target: Organized travel schedule and budget breakdown (250 words) with day-by-day technical activities and safety measures.

1. Question 1 : Write an engaging, well-organized Travel Plan and Budgetary Itinerary (250 words) for a 4-day student technical tour to an Aerospace and Satellite Launch Station, detailing transport logistics, budget calculations, planned technical visits, and safety protocols.

Answer:

4-Day Technical Tour to an Aerospace and Satellite Launch Station (Sriharikota) Transport logistics: A group of 40 students and 2 faculty members will travel in a hired AC bus from Chennai to Sriharikota (about 100 km) and return by the same bus.

Day 1: Leave college at 6.00 a.m., reach Sriharikota by 9.30 a.m. and complete the entry formalities and ID checks. Visit the Space Museum and the launch view gallery. Stay overnight in a nearby guest house.

Day 2: Attend a technical talk on launch vehicles and satellite orbits, and visit the rocket assembly and the launch pad areas as permitted by the officials.

Day 3: Visit the mission control centre to see how launches are monitored using telemetry, tracking and radar. Afternoon: a session on the rocket propulsion system and a Q&A with scientists.

Day 4: Morning visit to a nearby bird sanctuary at Pulicat Lake, then return to Chennai by 6.00 p.m.

Budget per student:

Bus (fuel, driver, toll): ₹1,500

Stay for three nights (shared): ₹1,800

Food (3 days): ₹1,200

Entry and permit charges: ₹300

Insurance and first aid: ₹200

Miscellaneous: ₹200

Total: n5,200

Safety protocols: Carry photo ID cards and a copy of the permission letter, follow the guides at all times, stay with the group during every visit, do not take photographs in restricted areas, wear closed shoes, drink enough water, and keep faculty contact numbers with you. Students with any health problem must inform the faculty before the journey.

2. Question 2 : Write a comprehensive Travel Plan and Budgetary Itinerary (250 words) for a 4-day student industrial exposure tour to a Major Commercial Port and Maritime Dry-Dock Facility. Include transport arrangements, per-student cost estimates (food, travel, stay), daily technical briefing stops, and industrial dockyard safety protocols.

Answer:

4-Day Industrial Exposure Tour to a Commercial Port and Dry-Dock Facility (Kochi)

Transport arrangements: Thirty-five students and two staff members will travel by train

(sleeper class) from Chennai to Ernakulam on the night of Day 1, and use a hired mini-bus for local travel in Kochi. The return journey will be on the night of Day 4.

Day 1: Depart Chennai in the evening by overnight train.

Day 2: Arrive in Kochi, check in at the hostel and attend a briefing on port operations. Visit the cargo terminal to observe cranes, container handling and the storage yard.

Day 3: Visit the shipyard and the dry-dock. Engineers will explain ship hull construction, repair, welding methods and how a ship is docked and drained. In the afternoon, a briefing on port logistics and customs.

Day 4: A harbour boat ride to see the navigation channel and port facilities. Departure by evening train.

Per-student budget:

Train fare (two-way): ₹1,300

Local transport: ₹600

Stay (two nights): ₹1,400

Food (4 days): ₹1,300

Entry, boat ride and insurance: ₹400

Total: n5,000

Dockyard safety protocols: Wear helmets, safety shoes and reflective jackets, stay within the marked walkways, keep away from cranes and moving vehicles, avoid the edge of the dock, never touch machinery or welding areas, and strictly follow the instructions of the safety officer. A first-aid kit and a faculty member will always stay with the group.

3. Question 3 : Write an engaging Travel Plan and Budgetary Itinerary (250 words) for a 4-day technical study tour to an Electric Locomotive Manufacturing Works and Automated Railway Signalling Center. Detail rail transit bookings, category-wise expenditure estimates, technical shop- floor visits, and site visitor safety rules.

Answer:

4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Centre

Rail transit bookings: Thirty students and two faculty members will travel by train in sleeper class. Tickets will be booked sixty days before the tour through the IRCTC website, and the group will use a hired mini-bus for local transport.

Day 1: Board the overnight train from Chennai. Students will complete a pre-tour worksheet on how electric traction works.

Day 2: Reach the destination city, check in at the guest house and visit the locomotive works.

The shop-floor visit covers the bogie section, the traction motor assembly line, the transformer winding shop, the welding bay and the final testing yard.

Day 3: Visit the automated railway signalling centre, where engineers will demonstrate the control panel, track circuits, interlocking systems and the automatic block signalling that controls train movement. Evening: a discussion with the engineers.

Day 4: A short visit to the railway museum, followed by the return journey by train. **Category-wise expenditure per student:**

Train tickets (two-way): ₹1,600

Stay (two nights): ₹1,400

Food: ₹1,200

Local transport: ₹500

Entry and permissions: ₹300

Total: n**5,000**

Site visitor safety rules: Wear helmets and safety shoes on the shop floor, walk only on marked paths, never cross railway tracks, do not touch overhead wires or equipment, keep a safe distance from moving locomotives, and follow the instructions of the guides. Mobile phones must be kept silent, and students must stay with the group at all times.

MODEL EXAM I – PRACTICE WORKSHEET

EN25C09: Professional English – I (Lab Integrated)

You Choose, We Do It
St. JOSEPH'S COLLEGE OF ENGINEERING
(An Autonomous Institution)
St. Joseph's Group of Institutions
OMR, CHENNAI - 119

PART - A (15 X 1 = 15 Marks)

Answer ALL questions. Each question carries 1 mark.

1. Fill in the blank with the correct form of the verb given in bracket:

The automated system _____ (monitor) the temperature continuously to ensure safe operation.

Answer: monitors

2. Fill in the blank with the correct past form of the verb given in bracket:

The engineers _____ (record) the vibration readings during yesterday's trial.

Answer: recorded

3. Complete the sentence with the appropriate comparative form of the adjective in bracket:

The newly developed composite material is significantly _____ (strong) than the conventional material while maintaining a lower weight.

Answer: stronger

4. Fill in the blank with an appropriate modal auxiliary verb: (must / can / might)

Engineers _____ follow safety regulations while operating heavy machinery.

Answer: must

5. Fill in the blank with a suitable modal auxiliary verb expressing capability: (can, may, might)

Modern sensors _____ detect even minor changes in temperature.

Answer: can

6. Combine the sentences into a single coherent sentence using a suitable relative pronoun:

(a) The laboratory developed a new testing device.

(b) The device measures material strength with high precision.

Answer: The laboratory developed a new testing device that measures material strength with high precision.

7. Complete the sentence using the correct form of the verb in bracket (First Conditional – If Clause):

If the temperature _____ (rise) above the permissible limit, the cooling system will activate automatically.

Answer: rises

8. Complete the sentence using the correct form of the verb in bracket (Third Conditional – Past Unreal Conditional):

If the technician had inspected the circuit earlier, the equipment _____ (not/fail) during operation.

Answer: would not have failed

9. Combine the following sentence using an infinitive of purpose (to + verb):

The mechanics installed dual damper springs. They wanted to isolate chassis resonance.

Answer: The mechanics installed dual damper springs to isolate chassis resonance.

10. Fill in the blank with the gerund form of the verb in bracket after the preposition:

The research team reduced energy consumption by _____ (optimize) the motor control algorithm.

Answer: optimizing

11. Convert the following active voice into the Impersonal Passive Voice:

The lab assistant poured 50 mL of hydrochloric acid into the titration flask.

Answer: It is observed that 50 mL of hydrochloric acid was poured into the titration flask.

12. Fill in the blank with the appropriate adverbial form derived from the adjective in bracket:

The robotic arm moves _____ (smooth) to ensure accurate positioning of the component.

Answer: smoothly

13. Rewrite the direct into indirect speech:

The project supervisor asked the engineer, "Did you calibrate the pressure sensor before testing?"

Answer: The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.

14. Identify the incorrect noun form and rewrite the sentence correctly:

The laboratory purchased several new equipments for conducting material testing.

Answer: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Fill in the blank using the correct irregular comparative form (better, worse, farther/further, less, more):

Because the heat sink had poor thermal paste contact, heat dissipation was _____ (bad) than expected.

Answer: worse

PART - B (10 × 2 = 20 Marks)

Answer ALL questions. Each question carries 2 marks.

1. Rewrite the following active operational and laboratory description in the Simple Past Tense:

The engineer calibrates the pressure sensor and verifies its accuracy.

Answer: The engineer calibrated the pressure sensor and verified its accuracy.

2. Draft two comparative sentences comparing a laptop and a desktop computer in terms of portability and screen size.

Answer: (Portability) A laptop is more portable than a desktop computer. (Screen Size) A desktop computer usually has a larger screen than a laptop.

3. Combine the technical statements into a single cohesive sentence using a defining or non-defining relative clause:

The engineer designed a new cooling system. It reduces the temperature of industrial machines.

Answer: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. Complete the following conditional sentences logically:

(a) If the battery voltage drops below the required level, _____

(b) If the technician had checked the connections earlier, _____

Answer: (a) the system will shut down automatically to prevent damage.

(b) the equipment might not have malfunctioned.

5. Convert the following active sentences into the Impersonal Passive Voice suitable for formal engineering reports:

The laboratory assistant heated the solution to 80°C before testing.

Answer: It is reported that the solution was heated to 80°C before testing.

6. Rewrite the following direct engineering query into Reported (Indirect) Speech:

The project manager asked the technician, "Have you inspected the pressure valves before starting the compressor?"

Answer: The project manager asked the technician whether he **had inspected the pressure valves before starting the compressor**.

7. Identify and correct the grammatical errors involving mass/uncountable nouns in the following sentences:

(a) The engineer received many feedbacks on the new design.

(b) The students received several useful advices from their teacher.

Answer: (a) The engineer received a lot of feedback on the new design.

(b) The students received several useful pieces of advice from their teacher.

8. Rewrite the sentences using an Infinitive of Purpose or a Gerund construction:

(a) The technician wears safety gloves. He wants to protect his hands.

(b) The cooling fan operates continuously. It prevents the motor from overheating.

Answer: (a) The technician wears safety gloves to protect his hands.

(b) The cooling fan operates continuously to prevent the motor from overheating.

9. Fill in the blanks with the correct adverbial forms of the bracketed words:

(a) The machine operates _____ (efficient) to reduce energy consumption.

(b) The technician checked the electrical connections _____ (careful) before starting the machine.

Answer: (a) efficiently

(b) carefully

10. Insert suitable modal auxiliary verbs to match the specified functional meaning:

(a) The robotic arm _____ lift heavy components with precision. [Capability]

(b) Visitors _____ not enter the testing area without authorization. [Prohibition]

Answer: (a) can

(b) must

PART - C (5 × 13 = 65 Marks)

Answer ALL questions (Choose either 'a' or 'b' where options are provided). Each question carries 13 marks.

Question 26

26. a) Write a set of eight clear, imperative safety instructions to promote safe and responsible use of the internet and protect users from cyber threats. (13 Marks)

Answer Details:

1. Use strong, unique passwords for every online account.
2. Enable two-factor authentication whenever it is available.
3. Never share passwords, one-time passwords, or sensitive personal information online.
4. Check website addresses and verify links before clicking them.
5. Do not open unexpected attachments or download files from unknown sources.
6. Keep your operating system, apps, and antivirus software updated.
7. Use secure networks and avoid sensitive transactions on public Wi-Fi.
8. Report suspicious messages, cyberbullying, and possible security breaches to a trusted adult or the appropriate authority.

Question 27 (Choose either 27.a or 27.b)

27. a) Write a descriptive essay of 250 words about "A Traditional art form in your region and how modern initiatives can preserve its cultural significance." Discuss its historical origins, traditional practices, and ways to promote and sustain it among younger generations. (13 Marks)

OR

27. b) Write a structured travel plan and descriptive feature (250 words) on "An Educational Trip to a

Place of Historical and Cultural Importance." Include travel arrangements, important landmarks, local food experiences, and observations of the region’s cultural heritage and traditions. (13 Marks)

Selected Option: 27.a

A Traditional Art Form and Its Cultural Significance

Bharatanatyam is a famous classical dance form from Tamil Nadu. It began in temples, where dancers performed stories about gods along with devotional music. Later it became a stage art, and today people perform it all over the world.

A Bharatanatyam performance has fast footwork, rhythm, hand gestures called mudras, and facial expressions. The dancers wear colourful silk costumes and jewellery. Music and instruments like the mridangam support the dance. Learning it takes years of practice and respect for the guru.

Many young people today are more interested in films and social media than in classical dance. To change this, schools and colleges can hold dance workshops and cultural clubs. Dance schools can offer cheaper classes and scholarships. Videos of performances and interviews with senior artists can be shared online. Local festivals can also give young dancers a chance to perform. Teachers should explain the meaning behind the movements, not just the steps.

In conclusion, Bharatanatyam is an important part of our culture and we should protect it. If we combine traditional training with modern tools, more young people will learn it and enjoy it.

Question 28 (Choose either 28.a or 28.b)

28. a) Write a formal Letter of Request to the Librarian of your college seeking permission to extend library working hours during the semester examination period. Explain the academic benefits, the

expected student participation, and the arrangements required to ensure proper use of library facilities. (13 Marks)

OR

28. b) Draft a formal Letter of Request to the Principal of your college seeking permission to participate in a technical workshop on Artificial Intelligence and Emerging Technologies organised by a reputed institution. Mention the objectives of participation, the expected learning outcomes, and the arrangements required. (13 Marks)

Selected Option: 28.b

From:

Harini G

First-Year B.E. Computer Science and Engineering St. Joseph's College of Engineering

Date: 19 October 2026

To:

The Principal

St. Joseph's College of Engineering

Respected Sir/Madam,

Subject: Request for permission to attend a workshop on Artificial Intelligence and Emerging Technologies – Reg.

I am a first-year B.E. Computer Science and Engineering student. I request permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies organised by a reputed institution.

I want to attend because I am interested in AI and would like to learn how it is used in real life. I will also get to see demonstrations and speak with experts. After the workshop, I hope to improve my technical knowledge and understand what career options are available in this field. This will help me choose projects in the future.

I will find out the schedule and registration details. I will make sure I do not miss important classes, and I will complete any missed assignments. After the workshop, I will submit my certificate and a short report to the department.

I kindly request you to grant me permission. Thank you for your time.

Yours faithfully,

Harini G

First-Year B.E. Computer Science and Engineering

Question 29 (Choose either 29.a or 29.b)

29. a) Compose an analytical essay (250–300 words) comparing the advantages and disadvantages of

Artificial Intelligence–Based Assessment versus Traditional Teacher-Based Assessment in engineering education. Evaluate assessment accuracy, feedback time, fairness, and the role of human judgement. (13 Marks)

OR

29. b) Write a compare-and-contrast essay of 250–300 words analysing the advantages and disadvantages of Living in Urban Areas versus Rural Areas. Discuss employment opportunities, cost of living, access to education and healthcare, and quality of life. (13 Marks)

Selected Option: 29.a

AI-Based Assessment versus Traditional Teacher-Based Assessment

Assessment helps students and teachers see how much has been learned. AI-based assessment uses software to check answers, quizzes, and code. In traditional assessment, a teacher checks the work. Both have good and bad points.

The biggest advantage of AI is speed. It can check objective questions quickly and give feedback at once, so students can find their mistakes and try again. It also uses the same rules for everyone, so marking is consistent. But AI can make mistakes if its data or rules are poor. It may not understand a correct answer written in an unusual way. If the system is biased, some students may be treated unfairly.

Teachers can judge more than the final answer. They can see how a student thinks, how creative the work is, and how the student performs in the lab. They can also explain mistakes in person. This is very useful for engineering projects. However, checking papers for a large class takes a lot of time, and a tired teacher may mark less carefully.

Neither method is perfect. AI is good for routine marking and quick feedback, while teachers are needed for complex work and for guiding students. A good solution is to use AI for first-level checking and let teachers review important results. There should also be a way for students to appeal their marks.

In conclusion, using AI together with human judgement can make assessment faster and fairer.

Question 30 (Choose either 30.a or 30.b)

30. a) Draft a personal email to your school friend sharing your experiences during the first semester of your engineering course. Describe your new friendships, classroom activities, adjustment to college life, and expectations for the upcoming semester examinations. (13 Marks)

OR

30. b) Draft a personal email to your school friend describing your experience attending the orientation programme for first-year engineering students at your college. Highlight the activities conducted, the introduction to academic facilities and college regulations, and how the programme helped you understand college life and prepare for your academic journey. (13 Marks)

Selected Option: 30.a

To: deebu09088@email.com

Subject: My First Semester of Engineering College

Dear Deebu,

How are you? I wanted to tell you about my first semester in engineering college.

College is very different from school, but I am enjoying it. I have made some good friends in my class. We have lectures, lab sessions, and a little programming. I am learning to manage my time and finish assignments on my own. A few subjects are tough, but I ask doubts and practise regularly, and it is helping.

Exams are coming soon, so I am revising my notes and solving practice questions. I made a study timetable too. I am a bit nervous, but I think I will be fine.

Write back and tell me about your college. Let’s meet during the holidays!

Your friend,

Harini